Context–Evidence Graph (CEG): In-Context Multi-Study Scientific Synthesis with Large Language Models

Author: `agent-kev` (https://github.com/agent-kev)
Contact: kev@openislet.org

Abstract

Objective. Insight in science often comes from seeing patterns, boundary conditions, and contradictions across hundreds or thousands of studies at once. That many full papers do not fit in a large language model's context window, and flat subject–predicate–object summaries are compact only because they discard the study arms, doses, subgroups, and experimental models that make findings comparable. This study presents the Context–Evidence Graph (CEG), a compact structured record of study findings designed so that a large, heterogeneous literature (or a compact view of it) can sit in a model's working context at once, with each finding's arms, effect size and direction, supporting and contradicting evidence, and experimental setting kept intact. The purpose is to let a model explore across studies, look for the variables that explain disagreements, reconcile conflicting results, and propose new hypotheses and concrete next experiments. Exploration at that scale needs a safeguard, because a fluent draft can cite real studies and quote real numbers while attaching a number to the wrong outcome, study arm, unit prefix, or inequality bound, or while carrying a result across an experimental or subgroup gap that the source study does not cover. CEG therefore pairs the representation with deterministic checks that run whenever a synthesis, hypothesis, or correction is written back to the shared graph. This paper formalizes the representation and evaluates that commit-time checking layer, stating what the checks guarantee and where checking of free-text sentences fails; it does not measure how much literature fits in a context window or how often exploration yields discoveries.

Materials and Methods. A CEG stores each reported finding as a structured record that links an intervention, population, outcome, measurement, and experimental setting to its source study (an `EvidenceAtom`). Each record names every study arm in the comparison (an explicit multi-arm contrast); states its measurement in one of five mutually exclusive formats, namely an exact value, a relative change, an interval, a censored bound, or a non-numeric outcome (a five-way disjoint-union quantity type); and is placed on a multi-dimensional map of experimental settings, such as mouse in vivo versus human tissue ex vivo versus clinical trial, rather than on a single 1-to-5 evidence ladder (a coordinate in a finite product partial order of contexts). Each cross-study conclusion is a claim that cites both the findings that support it and the findings that contradict it, and a proposal to carry a result across an experimental gap must name the follow-up experiment that would test it. Corrections made after extraction are kept in a reversible, checksum-verified edit log over exact-decimal JSON records (a state-guarded, invertible `JSONPatch` ledger, `RFC 6901`/`RFC 6902`, over canonicalized float-free JSON, `RFC 8785`). When anything is committed to the shared graph, a two-layer verifier checks six invariants, each computable in polynomial time: every reference resolves and every record matches the curated corpus in both directions (referential closure and bidirectional atom-to-corpus binding, I_1); the edit log replays forward and backward (bijective ledger replay, I_2); every stated number, unit, arm, and direction is grounded at three levels, namely Unicode-hardened lexical tokens (I_3^{tok}), depth-0 per-clause contrast and polarity rules (I_3^{clause}), and exact seven-coordinate typed propositions (I_3^{typed}); results move between experimental settings only where the declared ordering allows (partial-order transfer gating, I_4); nothing is promoted directly from an undigitized figure or an unverified source (a primary-source digitization gate, I_5); and supporting and counter-evidence stay disjoint, with claim attestations and locks on non-prose metadata (I_6). The study evaluates a zero-dependency Python reference implementation across exhaustive finite-domain self-tests, a constructive synthetic benchmark of 1,931 cases (1,521 adversarial, 410 valid) from 199 distinct templates—including 43 sentences from an independent adversarial pass and 48 gauntlet probes—and a primary-source evaluation on 12 peer-reviewed PubMed studies (31 `EvidenceAtom`s) spanning idiopathic pulmonary fibrosis, heart failure, and thoracic oncology. All of these experiments test the commit-time checks. A worked example (§5) carries ten findings from eight peer-reviewed studies on islet replacement for type 1 diabetes through extraction, classification against a clinical target setting, logged correction, and the decision on each candidate claim. A case study then applies the representation and the checks to a 12 November 2026 snapshot of the production corpus those studies come from, of 1,151 type 1

diabetes studies (§6.9): it reports the snapshot's size and structure and replays its 106-event curation ledger through the reference verifier with tamper controls.

Results. In brief, the formal guarantees held on every exhaustive test; the full verifier rejected every adversarial case in the core, stress, red-team, and gauntlet suites and in the real-text synthesis track; and the free-text lint showed its two predicted limits, namely wording errors it cannot catch and true sentences it refuses. In detail: (i) State-guarded ledger replay is deterministic, invertible, and tamper-evident (Proposition 1; 182 tampers tried across 15 kinds, 0 accepted), while Layer-1 structure-only verification (`check_top_digests=False`) rejects 26/26 structural tampers without consulting any precomputed digest. (ii) Direct quantitative promotion is sound relative to the declared context partial order, and no scalar evidence ranking is both sound and complete once incomparable models exist (Theorem 1; 0 disagreements across 26,344 ordered context pairs, and 0 faithful scalar rankings among 46,656 assignments). (iii) Any surface prose verifier reading only citations, numerals, units, and polarity cues returns identical verdicts on a true pooled-citation sentence and its endpoint-misbound twin (Theorem 2; Corollary 2.1 verified across 242/242 canonical renderings). (iv) Typed proposition verification accepts a unique proposition per atom up to arm-set ordering and ASCII case (Theorem 3; 133/133 single-coordinate mutants rejected). Across synthetic suites, the full `V3` verifier rejects 600/600 core, 600/600 stress, 43/43 red-team, and 38/38 gauntlet adversarial cases while accepting 200/200 core and 10/10 gauntlet valid controls, and rejects 1,161/1,161 record-level adversarial controls outside the residual suite under a Mode B self-attesting adversary that recomputes claim attestations after tampering (60/60 non-prose metadata tampers rejected). On stress valid controls it accepts 100/200 (refusing pooled citations and unmodeled derived or dispersion numerals), and on the residual open-class prose suite it rejects 20/240 (catching pooled `R1` headers while leaving 220/240 single-tuple endpoint renamings and unlisted lexical inversions open as predicted). On the 12-study PubMed evaluation, `V3` accepts 6/36 and refuses 30/36 verbatim abstract sentences (most often because of inline confidence intervals, hazard ratios, event denominators, and publisher thin spaces), while accepting 24/24 valid grounded synthesis records and rejecting 60/60 adversarial clinical synthesis defects in both normal and Mode B self-attesting evaluation. At production scale (1,151 studies, 14,105 outcomes), the reference verifier accepts the complete 106-event curation ledger and rejects 5/5 structural tampers of it without consulting any digest; in this snapshot none of the outcomes is certified against its primary source, and the settled corpus occupies 20,412,758 bytes of canonical JSON (§6.9).

Conclusion. A CEG is a way to put a large structured literature in front of a language model at once and let it explore freely, while a deterministic gate checks what is written back. The gate gives exact guarantees for structured (typed) claims, for reversible edit logs, and for transfers between experimental settings on the declared context map, which is where swapped arms, unit mix-ups, and unstated leaps such as mouse-to-human are stopped. Free-text sentences are only linted, conservatively, with measured blind spots and a measured refusal rate on raw clinical prose, so decisions should rest on the typed claims. The experiments establish this integrity layer as a foundation for discovery workflows. A production corpus of 1,151 studies shows the representation and its correction log operating at scale, and shows what remained open at the time of the snapshot: its outcomes were not certified against primary sources, and its full serialization is of the order of millions of tokens. How well exploration over a CEG generates sound hypotheses remains to be evaluated.

Keywords: AI for science; hypothesis generation; in-context multi-study synthesis; evidence graphs; translational research; claim verification; provenance

1. Introduction

Scientific breakthroughs often come from seeing patterns, boundary conditions, and contradictions across hundreds or thousands of studies at once: an effect that appears only at one disease stage, a treatment that works in mice but not in human tissue, two trials that disagree because they enrolled different patients. Large language models (LLMs) can reason over whatever is placed in their context window, but a literature rarely fits there in a usable form. Full papers are long, so even a modest collection exceeds or clutters the window, while flat knowledge-graph triples of the form subject–predicate–object are compact but strip away the study arms, doses, subgroups, and experimental models that make two findings comparable. Context–Evidence Graphs (CEGs) are designed to sit between these extremes. Quantitative evidence in biomedicine and engineering comes from heterogeneous sources: cell assays, animal disease models, explanted human tissue, subgroup analyses, and randomized clinical trials. A CEG condenses each reported finding into a small structured record that keeps its study arms, measurement unit, statistic, sample size, and experimental

context, and it records each cross-study conclusion together with the findings that support it and the findings that contradict it. The aim is that a large, heterogeneous body of evidence, or a compact view of it, can be loaded into a model's working context at once, so that the model can compare results across studies, look for the variables that explain disagreements, and propose new hypotheses together with the experiments that would test them.

To see both the opportunity and the hazard in plain terms, consider synthesizing thousands of studies on preserving or replacing insulin-producing beta cells in type 1 diabetes. Across that literature, candidate therapies routinely reverse high blood glucose in newly diabetic mice, whereas experiments on donated human pancreatic islets kept alive in culture test genuine human cells without a bloodstream or an active autoimmune attack, and clinical trials span newly diagnosed children who retain residual insulin production through adults with longstanding disease receiving systemic immunosuppression. Distilling each paper into compact, structured study arms, doses, patient groups, follow-up windows, and measured outcomes allows a language model to hold thousands of structured findings in working memory at once—comparing results across an entire literature to notice, for example, whether an intervention succeeds only before beta-cell loss is complete or reverses direction under a particular immunosuppressive regimen. Yet at that scale, a fluent synthesis that swaps active and control C-peptide values, confuses a micromolar culture concentration with a clinical dose, or presents mouse reversal as established human clinical efficacy states a substantive falsehood while citing real papers and real numbers. (§5 works through ten findings from eight peer-reviewed studies on exactly this problem, §6 evaluates the framework formally and empirically on synthetic benchmarks and 12 peer-reviewed clinical and preclinical studies across idiopathic pulmonary fibrosis, heart failure, and thoracic oncology, and §6.9 reports the production corpus of 1,151 type 1 diabetes studies from which §5's findings are drawn.)

Why large-scale in-context discovery needs a deterministic commit gate. Exploration at this scale produces drafts: syntheses, hypotheses, and corrections that a model proposes to write back to the shared graph. LLMs and retrieval-augmented generation (RAG) pipelines draft fluent quantitative syntheses rapidly [1–3], and their surface citations and numerals often appear genuine: the cited papers exist and the quoted numbers occur somewhere in those papers. Auditing whether each number remains bound to the right outcome, arm, unit scale, inequality direction, and target population is slow, and automated factuality evaluators largely leave that structural binding unchecked. If such drafts are saved unchecked, each error becomes part of the evidence that the next round of exploration reads. The characteristic errors recombine true material from cited papers rather than inventing citations outright:

1. **Mixing numbers across outcomes or across cited studies.** A study reports a target biomarker reduction (29.4 ng/mL) alongside a slice viability control (96.2%), or one sentence cites three trials in sequence, clause by clause. Attaching 96.2 to the biomarker, rotating values among the three cited clauses ($A \to B \to C \to A$), or copying one clause's number into a second clause while keeping it in the first all pass a check that only asks whether each number appears somewhere in the cited study or sentence (a study-wide or sentence-wide bag-of-numbers check).
2. **Swapping treatment and control arms or misreading negated bounds.** The pair of numbers $(29.4, 48.6)$ is kept but assigned to the wrong arms, so the treatment value is reported for the comparator and the comparator value for the treatment. The same swap can hide inside a comma-separated aside (an appositive), as in *"the placebo arm had a change, −28.5 mL, compared with treatment"*, where the number between the commas belongs to the treatment arm but reads as the placebo's. Or a negated bound is read without flipping it: `not < 5%` means at least 5%, and `never > 28 days` means at most 28 days.
3. **Confusing look-alike units, symbols, or figure labels.** Simple pattern-matching number extractors (regular expressions) miss numbers written as words (*tripled*, *sixty-four percent*) and as fraction characters (vulgar fractions); are fooled by invisible or combining Unicode characters that split a number (`U+0301`, `U+20DD`) and by Cyrillic or Greek letters that look identical to Latin ones (homoglyphs) [30, 45]; merge micromolar concentration (`uM`, `mM`, `nM`) with micrometre or millimetre length (`um`, `mm`, `nm`) when letter case is ignored; or mistake a figure panel label (`Fig. 28 d`) for a duration (`days`).
4. **Confusing relative percentages with raw values, or dropping third study arms.** Forcing every outcome into two slots, `(treatment, control)`, turns a 72.0% relative reduction into what looks like a measured level and silently drops the active comparator arm of a three-arm study.
5. **Flattening different experimental models onto a single ranking ladder.** Evidence hierarchies with a single axis rank study designs on one ordinal scale [4]. Translational and clinical contexts vary

along multiple axes simultaneously. A murine bleomycin model provides systemic pharmacokinetics and immune cell recruitment, whereas human precision-cut lung slices (PCLS) preserve aged human target biology and fibrotic matrix ex vivo without systemic circulation [5, 6]; similarly, a stage II–IIIA resected lung cancer primary analysis and an overall stage IB–IIIA population differ along a disease-stage axis even within one Phase 3 trial [43]. Contexts that cannot be ordered against each other (incomparable contexts) admit no single scalar ranking without either discarding complementary evidence or licensing unsupported extrapolation.

6. **Unrecorded post-extraction edits or unverified claims tucked into metadata.** Raw literature extractions need repair after the fact: fixing case-folded units, restoring dropped comparator arms, correcting subgroup sample sizes, or flagging Kaplan–Meier estimates read off a figure that has not been digitized. Without a reversible, checksum-guarded edit log (an invertible, digest-guarded patch ledger) and locks over the metadata fields that are not prose, edits cannot be traced back to the raw extraction, and unverified statements can be placed in fields that no check reads.

CEGs address these failure classes with one representation and one set of deterministic checks: the structured records that the model explores are the same records that the checks run against (formally, a typed hypergraph with a deterministic verification pipeline). The central design principle is a strict separation between **exact structural guarantees** and **heuristic prose linting**. A free-text sentence cannot be tied to one piece of evidence just by inspecting its surface citations, numerals, units, and polarity cues; Theorem 2 proves an exact indistinguishability barrier for that class of verifiers. CEGs therefore place every decision-bearing claim in a typed proposition, a structured statement that is compared field by field, on seven coordinates, with its `EvidenceAtom` (Theorem 3); govern movement between experimental settings by a product partial order (Theorem 1); track corrections in an edit log that replays in both directions (bijective `JSONPatch` replay, Proposition 1); and pair these exact layers with a conservative prose lint whose false-negative boundaries and verbatim clinical refusal rate are measured directly in §6.

Design intent: explore free in context, verify at commit. CEGs are meant to be used in two regimes, and the distinction governs how every result in this paper should be read. *Regime 1 (in-context exploration and hypothesis discovery) is the primary operating mode.* Condensing heterogeneous studies into compact `EvidenceAtom`s, with their study-arm variables, effect magnitudes and directions, and positions on the multi-axis map of experimental settings (`ContextSignature` coordinates), lets an entire multi-study outcome network sit inside a language model's context window at once. While exploring, synthesizing, and generating hypotheses, the model is intended to traverse that network freely in natural language, under no prose-lint restriction: spotting cross-study patterns, looking for moderating variables where effects differ between a claim's supporting and counter-evidence findings (T^+ versus T^-), proposing cross-model hypotheses, and drafting candidate graph additions or curation edits.

Regime 2 (commit-time verification) is its guardrail. Verification under I_1–I_6 sits only at the persistence and promotion boundary. It checks what is committed back to the shared graph: claim records (`SynthesisClaimRecord`s), their structured field-by-field statements (`TypedProposition`s) and evidence-anchor sentences (`prose_sentences`), and curation edits (`ReversiblePatchEvent`s). At that boundary I_3 catches swapped numbers, arms, and units, and I_4 sorts cross-setting claims into three kinds. A result whose source setting already covers the target (`DIRECT_MATCH`) may be promoted directly. A hypothesis that bridges a gap between comparable settings (a `CANDIDATE_UPGRADE` over a `BOUNDED_BRIDGE`) must carry a concrete follow-up experiment that would test every unresolved experimental gap (a `ProspectiveFalsifierGate` covering every unresolved axis). An explanation in terms of a moderating variable is recorded as a `SCOPING_QUALIFICATION`. The gate does more than tag cross-context claims: a `SUPPORTS` or `CANDIDATE_UPGRADE` record is rejected with `I4_INADMISSIBLE_CONTEXT_TRANSFER` whenever any of its supporting tuples is an `INADMISSIBLE_TRANSFER` (§4.1). The guardrail serves discovery: it keeps the shared graph clean as it grows and leaves exploration itself unrestricted.

This two-regime reading is a statement of design intent, not an empirical result. This paper formalizes the CEG representation and empirically benchmarks its Regime 2 integrity and transfer layer as the foundation for large-scale discovery workflows: every experiment in §6 evaluates the commit gate (Regime 2) only, and no experiment in this paper measures how much literature fits in a context window, the quality of upstream exploration, or the rate at which it yields discoveries.

Contributions

1. **A compact evidence-graph representation and two-layer verification model (§3–§4).** Evidence atoms with a strict five-way disjoint-union quantity type, truncation-free multi-arm contrasts, colon-safe bibliographic tuple identifiers (`PMID:...`, `DOI:...`), product partial-order transfer gating, invertible state-guarded `JSONPatch` ledgers, non-prose surface digest locks, and two-layer verification separating structure-only closure (`Layer 1`) from cryptographic digest pinning (`Layer 2`).
2. **Formal guarantees and impossibility bounds (§4.3).** Proofs of ledger determinism, invertibility, and tamper evidence (Proposition 1); transfer soundness and scalar-ranking impossibility on incomparable context posets (Theorem 1); surface-verifier indistinguishability on pooled citations and canonical single-tuple completability (Theorem 2 and Corollary 2.1); and unique seven-coordinate exactness of the typed proposition layer (Theorem 3).
3. **Zero-dependency reference implementation, dual-mode synthetic benchmark, and primary-source PubMed evaluation (§5–§6).** A standard-library implementation (`ceg_reference.py v3.1`, byte-identical across Python `3.9`–`3.14` under both normal and `-O` optimized execution), exhaustive finite-domain self-tests, a 1,931-case synthetic benchmark evaluated under both raw tamper (`Mode A`) and self-attesting tamper (`Mode B`), and an empirical evaluation on 12 peer-reviewed PubMed studies across idiopathic pulmonary fibrosis (IPF), heart failure (HF), and thoracic oncology [33–44].
4. **A worked example on real studies (§5) and a production-scale case study (§6.9).** Ten findings from eight peer-reviewed studies on islet replacement for type 1 diabetes, curated into the representation, classified against a clinical target context, and submitted to the gate, with four cross-study hypotheses held behind prospective falsifier gates and four adversarial probes rejected. A dated snapshot of a corpus of 1,151 type 1 diabetes studies (14,105 outcomes) with its size and structure; a replay of its 106-event correction log through the reference verifier, with tamper controls; and a plain account of what that snapshot does not verify (no certified outcomes, a prioritized subset, language-model extraction, and a serialization of the order of millions of tokens).

2. Related Work

2.1 Evidence grading, indirectness, and translational validity

Systematic reviews follow PRISMA reporting standards [8] and Cochrane synthesis methodology [9], while GRADE evaluates certainty of evidence across risk of bias, inconsistency, indirectness, imprecision, and publication bias [7]. Within GRADE, indirectness captures mismatches across population, intervention, comparator, and outcome (PICO) between source studies and target clinical questions [10]. Critiques of rigid evidence pyramids emphasize that study design alone does not determine validity [4]. In preclinical-to-clinical translation, animal efficacy frequently fails to replicate in human trials [11, 31, 32]; American Thoracic Society consensus guidance on pulmonary fibrosis models explicitly notes that murine bleomycin fibrosis resolves spontaneously and recommends complementary confirmation in human fibrotic tissue such as precision-cut lung slices [5, 6]. CEGs do not replace statistical meta-analysis or clinical judgment; instead, they encode indirectness formally as a product partial order $(\mathcal{C}, \preccurlyeq_{\mathcal{C}})$, compute the exact set of unresolved axes $U(c_s, c_t)$, and block quantitative parameter promotion across unresolved or below-floor gaps.

2.2 Computational models of scientific claims, argumentation, and provenance

Computational claim models build on Toulmin's argument layout [12] and Dung's abstract argumentation frameworks [13]. Nanopublications link assertions to provenance graphs [14], and micropublications model attribution and support/challenge networks over biomedical statements and data figures [15]. W3C `PROV-DM` [16] and the FAIR data principles [17] standardize lineage metadata, while large biomedical knowledge graphs such as SemMedDB [18] and PrimeKG [19] aggregate subject–predicate–object triples across literature and ontologies. Because entity–relation triples omit multi-arm contrasts, exact measurement units, sample sizes, and experimental context coordinates, they cannot detect arm swaps, unit-scale folds, or subgroup-to-overall promotions. CEGs operate one level deeper, representing individual study outcomes as typed hypergraph atoms bound to a replayable corpus extraction.

2.3 Automated factuality and table/claim verification

Attribution frameworks such as AIS [1], FActScore [2], and ALCE [3] decompose generated text into atomic statements and score support using natural-language inference (NLI) models. In biomedical and

scientific verification, SciFact checks single-sentence claims against abstracts [20], Evidence Inference classifies intervention–comparator–outcome directions in randomized trial reports [21], RobotReviewer extracts risk-of-bias judgments [22], and TabFact [23] and SciTab [24] benchmark compositional reasoning over tables. Neural entailment models handle open-ended lexical paraphrase that deterministic parsers refuse, but their verdicts are probabilistic, model-dependent, and vulnerable to subtle numerical, unit-case, or subgroup swaps. CEG verification is complementary and strictly deterministic: upstream human curators or language models extract structured records and draft claims, after which the CEG verifier enforces exact algebraic, numeric, unit, and context invariants.

2.4 Canonical JSON, document patching, and Unicode security

Post-extraction curation in CEGs builds on IETF `RFC 6901` JSON Pointer [25], a state-guarded restriction of `RFC 6902` JSON Patch [26], and `RFC 8785` JSON Canonicalization Scheme (JCS) [27] restricted to float-free trees so decimal quantities never pass through IEEE-754 binary floating-point conversion. Lexical normalization and homoglyph defenses draw on Unicode Technical Report #36 (*Unicode Security Considerations*, Revision 15) [30] and its active normative companion, Unicode Technical Standard #39 (*Unicode Security Mechanisms*, Version 18.0.0) [45].

3. Formal Model

Informally, a CEG represents each reported study finding as a self-contained, unit-aware building block (`EvidenceAtom`), locates its experimental setting on a multi-axis map of contexts rather than a single ranking ladder, records every post-extraction correction as a reversible edit, and represents each cross-study conclusion as a hyperedge connecting supporting and opposing findings. Three of these pieces do more than block errors. A claim's supporting and counter-evidence anchors (T^+ and T^-) record where studies disagree, which is how the graph represents scientific anomalies. The multi-axis context map (`ContextSignature`) records the setting in which each finding holds, which is how it represents boundary conditions. The `ProspectiveFalsifierGate` records a concrete next experiment (which arms to compare head to head, on which readout, against what pass threshold) for testing a hypothesis that crosses experimental models. Formally, all CEG structures are finite and serialize to canonical float-free JSON (`RFC 8785`, rejecting `NaN` and `±Infinity` on parse, and integers too large for a double-precision number to hold exactly). Continuous and discrete measurement magnitudes are stored as exact ASCII decimal strings and compared via arbitrary-precision `Decimal` arithmetic.

Key terms in plain language. The definitions below use the following names, which are also the names used in the reference implementation.

- **`EvidenceAtom`**: one reported finding from one study, stored with its arms, numbers, unit, statistic, sample size, experimental setting, and a pointer to the source passage.
- **`ContextSignature`**: the experimental setting of a finding or of a claim, given as one value on each axis of the context map (for example model system, disease stage, dosing, and endpoint).
- **Transfer class**: the answer to the question of whether a finding from one setting can speak to another. It is `DIRECT_MATCH` when the source setting covers the target, `BOUNDED_BRIDGE` when a gap remains that a named experiment could close, and `INADMISSIBLE_TRANSFER` when the gap cannot be bridged.
- **`SynthesisClaimRecord`**: one cross-study conclusion or hypothesis, together with the findings that support it (T^+) and the findings that count against it (T^-).
- **`TypedProposition`**: the structured, field-by-field form of what a claim states about one finding. The exact checks compare this form.
- **Evidence-anchor sentences (`prose_sentences`)**: the free-text sentences attached to a claim for human readers. They are linted conservatively and are not verified exactly.
- **`ProspectiveFalsifierGate`**: the follow-up experiment attached to a bridging hypothesis, naming the arms to compare, the readout, and the pass threshold.
- **`ReversiblePatchEvent`**: one logged correction to the extracted data, which can be replayed forward or undone.
- **Invariants I_1–I_6 and the two layers**: the six conditions that a commit must satisfy (§4.1). Layer 1 checks the content itself, and Layer 2 adds comparisons against stored checksums.

3.1 Product context partial order and transfer classifier

Let $D = \{1, ..., d\}$ index experimental or clinical validity axes (for example, *model system*, *disease chronicity or stage*, *dosing and exposure*, and *endpoint maturity*).

In plain terms: each experimental setting is a point on several independent axes (for example model system, disease stage, dosing, and endpoint); a result may be carried directly from one setting to another only when the source setting covers the target on every axis, and otherwise the definition says exactly which axes remain open and whether the gap can be bridged at all.

Definition 1 (product context poset and transfer classifier). For each axis $k \in D$, let $(\mathcal{C}_k, \preccurlyeq_k)$ be a finite partially ordered set (reflexive, antisymmetric, and transitive), where $x \preccurlyeq_k y$ encodes the explicit domain judgment that condition y dominates or covers condition x toward deployment validity on axis k. Incomparable elements $x \parallel_k y$ represent complementary experimental models or non-nested patient strata. The product context space is

$$\mathcal{C} = \prod_{k=1}^{d} \mathcal{C}_k, \qquad c \preccurlyeq_{\mathcal{C}} c' \Leftrightarrow \forall k \in D : c_k \preccurlyeq_k c'_k.$$

Given a source context $c_s \in \mathcal{C}$ and a target context $c_t \in \mathcal{C}$, the *unresolved axis set* is

$$U(c_s, c_t) = \{\, k \in D : c_{t,k} \not\preccurlyeq_k c_{s,k} \,\}.$$

Elements $x, y \in \mathcal{C}_k$ have a *common upper bound*, written $\mathrm{cub}_k(x, y)$, iff $\exists z \in \mathcal{C}_k$ such that $x \preccurlyeq_k z$ and $y \preccurlyeq_k z$. Let $b : B \to \bigcup_{k \in B} \mathcal{C}_k$ assign to a subset of axes $B \subseteq D$ a minimum *bridge floor* $b(k) \in \mathcal{C}_k$. The three-way transfer classifier $\tau_b : \mathcal{C} \times \mathcal{C} \to \{\texttt{DIRECT_MATCH}, \texttt{BOUNDED_BRIDGE}, \texttt{INADMISSIBLE_TRANSFER}\}$ is

$$\tau_b(c_s, c_t) = \begin{cases} \texttt{DIRECT_MATCH} & \text{if } U(c_s, c_t) = \varnothing, \\ \texttt{INADMISSIBLE_TRANSFER} & \text{if } \exists k \in U(c_s, c_t) : \neg\, \mathrm{cub}_k(c_{s,k}, c_{t,k}) \vee \big(\, k \in B \wedge b(k) \not\preccurlyeq_k c_{s,k} \,\big), \\ \texttt{BOUNDED_BRIDGE} & \text{otherwise.} \end{cases}$$

`DIRECT_MATCH` holds iff $c_t \preccurlyeq_{\mathcal{C}} c_s$ on every axis. `BOUNDED_BRIDGE` holds when one or more axes remain unresolved ($U \neq \varnothing$), yet on every unresolved axis $c_{s,k}$ shares a common upper bound with $c_{t,k}$ and clears any declared bridge floor $b(k)$, permitting conditional synthesis behind an explicit prospective falsifier gate. `INADMISSIBLE_TRANSFER` holds whenever an unresolved axis lies below its bridge floor or on a disconnected branch with no common upper bound. Poset axioms and floor membership are checked algorithmically as part of graph integrity (I_1).

3.2 Disjoint-union quantities, multi-arm contrasts, and colon-safe evidence atoms

In plain terms: every measurement is written in exactly one of five formats (a value with its unit, a relative change with its direction, an interval, a bound such as more than 28 days, or an explicit statement that no number is available), and the fields of the other four formats must be empty, so a percentage change can never be mistaken for a measured level.

Definition 2 (strict disjoint-union quantity type). Every quantitative slot takes a value in the five-variant disjoint union $\mathcal{Q} = \mathcal{Q}_{\text{level}} \sqcup \mathcal{Q}_{\text{rel}} \sqcup \mathcal{Q}_{\text{int}} \sqcup \mathcal{Q}_{\text{cens}} \sqcup \mathcal{Q}_{\text{miss}}$, where every inactive variant's fields must be `None` and every literal v must be a canonical ASCII decimal matching `-?(?:0|[1-9][0-9]*)(?:\.[0-9]+)?` in full (an optional minus sign, no plus sign, no leading zero, no exponent, ASCII digits only):

- **`point_level`**: (ρ, v, u, d_u) with $\rho \in \{=, \approx\}$, non-empty unit u, non-empty unit dimension d_u, and `lower = upper = change_direction = missing_reason = None`.
- **`relative_change`**: (ρ, v, u, δ) with ρ any of the six relation operators of `RELATION_OPS` (`=`, `≈`, `<`, `<=`, `>`, `>=`), $u \in \{\%, \texttt{fold}, \texttt{ratio}, \texttt{log2_fc}\}$, mandatory direction $\delta \in \{\texttt{increase}, \texttt{decrease}, \texttt{neutral}\}$, and `lower = upper = missing_reason = None`.
- **`interval_bound`**: $([v_{\text{lo}}, v_{\text{hi}}], u, d_u)$ with $v_{\text{lo}} \leq v_{\text{hi}}$ as exact `Decimal`s, non-empty u and d_u, and `relation = value = change_direction = missing_reason = None`.
- **`censored_observation`**: (ρ, v, u) with strict inequality or bound operator $\rho \in \{>, \geq, <, \leq\}$, non-empty u, and `lower = upper = change_direction = missing_reason = None`.
- **`missing_numeric`**: (`missing_reason`) with non-empty reason string and `relation = value = lower = upper = unit = unit_dimension = change_direction = None`.

In plain terms: a comparison must list every study arm that the source reports for that outcome, so the third arm of a three-arm trial cannot be silently dropped, and an `EvidenceAtom` bundles one outcome's comparison, values, statistic, sample size, experimental setting, and pointer back to the source passage into a single record.

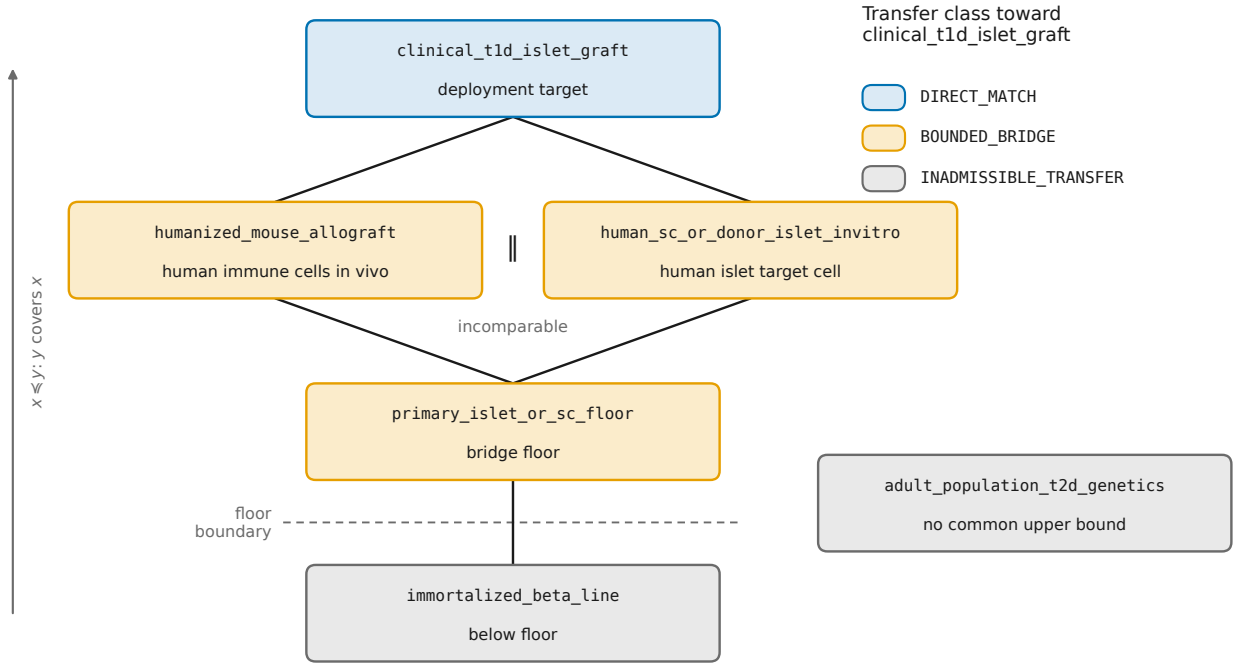


Figure 1. Hasse diagram of the six-element `model_system` *axis in the worked type 1 diabetes example (§5).* `humanized_mouse_allograft` *and* `human_sc_or_donor_islet_invitro` *are incomparable (‖), both clear* `primary_islet_or_sc_floor`*, and both share* `clinical_t1d_islet_graft` *as a common upper bound (*`BOUNDED_BRIDGE`*), whereas* `immortalized_beta_line` *sits below the floor and* `adult_population_t2d_genetics`*, a population genetic association in adults without a transplant, shares no upper bound with the target (*`INADMISSIBLE_TRANSFER`*).*

Definition 3 (truncation-free multi-arm contrast and `EvidenceAtom`). Let $\mathcal{V}(s)$ denote the declared arm variables of study s, and $\mathrm{Mentions}(t) \subseteq \mathcal{V}(s)$ the arm variables named in the primary source for outcome tuple t. A *multi-arm contrast* $\gamma = (A_{\mathrm{focal}}, A_{\mathrm{comp}}, \kappa)$ over duplicate-free tuples of arm IDs satisfies: (1) $A_{\mathrm{focal}} \neq \varnothing$, $A_{\mathrm{focal}} \cup A_{\mathrm{comp}} \subseteq \mathcal{V}(s)$, and $A_{\mathrm{focal}} \cap A_{\mathrm{comp}} = \varnothing$; (2) cardinality agreement with $\kappa \in \{\texttt{single_arm}, \texttt{pairwise}, \texttt{multi_arm_joint}\}$ ($(|A_{\mathrm{focal}}|, |A_{\mathrm{comp}}|) = (1,0)$ for `single_arm`, $(1,1)$ for `pairwise`, and $|A_{\mathrm{focal}}| + |A_{\mathrm{comp}}| \geq 3$ for `multi_arm_joint`); and (3) *non-truncation*: $\mathrm{Mentions}(t) \subseteq A_{\mathrm{focal}} \cup A_{\mathrm{comp}}$.

Outcome tuple identifiers take the right-anchored form $t = s : o$, partitioned at the **rightmost** colon (`split_outcome_tuple`, `rsplit(":", 1)`), where the outcome local identifier o is the non-empty remainder and therefore contains no colon; to be recognized as a citation in prose (`TUPLE_RE`) it must have the form `O\d+[A-Za-z0-9_-]*` (`O1`, `O2_itt`). The study/document prefix s may carry standard bibliographic namespace colons and slashes (`S104`, `PMID:24836310`, `DOI:10.1056/NEJMoa1402582`). An `EvidenceAtom` is the 13-tuple

$$a = (\ \texttt{ATOM:}\ t, t, s, \mathrm{ep}, c_a, \gamma, q_{\mathrm{focal}}, q_{\mathrm{comp}}, \mathrm{stat}, n, p_{\mathrm{int}}, \pi, \mathrm{prov}\ ),$$

with balanced-bracket endpoint name ep free of forbidden Unicode, context signature $c_a \in \mathcal{C}$, $q_{\mathrm{focal}} \in \mathcal{Q}$, $q_{\mathrm{comp}} \in \mathcal{Q} \cup \{\perp\}$, statistic kind stat from the closed vocabulary `STATISTIC_KINDS` (`mean`, `median`, `proportion`, `rate`, `rate_ratio`, `hazard_ratio`, `odds_ratio`, `difference`, `fold_change`, `qualitative`), non-negative integer sample size $n \in \mathbb{Z}_{\geq 0} \cup \{\perp\}$ (`bool` and `float` rejected), optional interval/p-value string p_{int}, closed polarity $\pi \in \{\texttt{favorable}, \texttt{unfavorable}, \texttt{neutral_or_mixed}, \texttt{unresolved}\}$ (`ATOM_POLARITIES`), and `ProvenancePin` prov = (doc_id, loc, quote, `requires_figure_digitization`, `primary_pdf_verified`). Qualitative outcomes require `missing_numeric`, and outcomes with `requires_figure_digitization = True` cannot carry a `point_level` quantity at all (neither $=$ nor $\approx$).

3.3 Hypergraph claims, non-prose surface locks, and bijective patch ledger

In plain terms: the graph consists of the context map, the studies, their arms, the findings, and the claims; each claim is a record that points to the findings supporting it and the findings against it, says what kind of claim it is and which setting it is about, and carries both its free-text sentences and its field-by-field structured statements.

Definition 4 (Context–Evidence Graph and synthesis claim hyperedge). A CEG is $\mathcal{G} = (\Lambda, \mathcal{S}, \mathcal{V}, \mathcal{A}, \mathcal{H})$. Each synthesis claim $h \in \mathcal{H}$ is a directed hyperedge

$$h = \big(\ \mathrm{id}, \omega, c_{\mathrm{target}}, T^{+}, T^{-}, \Sigma_{\mathrm{prose}}, \mathcal{P}_{\mathrm{typed}}, g_{\mathrm{prosp}}, \mathrm{prom}, \alpha\ \big),$$

connecting supporting tuples T^{+} and counter-evidence tuples T^{-} to a claim with closed epistemic polarity $\omega \in \{$`SUPPORTS`, `CONTRADICTS`, `SCOPING_QUALIFICATION`, `CANDIDATE_UPGRADE`, `FALSIFIED_GRAVEYARD`$\}$ at target context $c_{\mathrm{target}} \in \mathcal{C}$, prose sentences Σ_{prose}, typed propositions $\mathcal{P}_{\mathrm{typed}}$, optional `ProspectiveFalsifierGate` g_{prosp}, promotion request flag $\mathrm{prom} \in \{\mathrm{true}, \mathrm{false}\}$, and optional deterministic claim-attestation digest α (the field `attestation_sha256`, computed by `compute_claim_attestation_sha256` over the record's structured fields, cited tuples, and extracted numeric mentions). These ten components are the ten fields of `SynthesisClaimRecord`. When a record is serialized as an envelope (`record_to_envelope`), every key outside `ACTIVE_PROSE_KEYS = {"prose_sentences"}` is canonicalized and bound into the non-prose surface digest $H_{\mathrm{meta}} = H(h \setminus \Sigma_{\mathrm{prose}})$ (`non_prose_surface_sha256`). H_{meta} is not a field of the record: it is held beside the record and supplied to the verifier as the expected value, so unverified claims cannot be smuggled inside auxiliary metadata strings.

In plain terms: every correction made after extraction is logged as one small edit to one study's record, with the old value, the new value, a reason, and a checksum of that study's record before and after, so the whole history can be replayed forward to the curated data or backward to the raw extraction.

Definition 5 (state-guarded bijective `JSONPatch` ledger). Let D_0 be the raw extracted corpus JSON and D_m the settled corpus JSON. A curation ledger $\mathcal{L} = (e_1, ..., e_m)$ is a sequence of `ReversiblePatchEvents`

$$e_i = (\mathrm{id}_i, s_i, \mathrm{ptr}_i, v_i^{\mathrm{old}}, v_i^{\mathrm{new}}, \mathrm{rat}_i, H_i^{\mathrm{pre}}, H_i^{\mathrm{post}})$$

satisfying: (a) ptr_i is a canonical `RFC 6901` JSON Pointer (array indices match `0` or `[1-9][0-9]*`, forbidding `-1` or leading-zero aliases) addressing an existing mutable target strictly inside study s_i's record and never altering `study_id`; (b) $v_i^{\mathrm{old}} \neq v_i^{\mathrm{new}}$ under canonical `RFC 8785` encoding (forbidding no-op padding); (c) event IDs id_i are pairwise distinct, and each event's identifier, study identifier, pointer, and rationale carry no forbidden Unicode (`I2_FORBIDDEN_UNICODE_IN_LEDGER_STRING`); and (d) H_i^{pre} and H_i^{post} equal the `RFC 8785` SHA-256 digests of study s_i's isolated JSON subtree immediately before and after replacing v_i^{old} with v_i^{new}.

4. Verification Invariants, Formal Guarantees, and Prose Lint Boundaries

The checks in this section run only when something is committed to the shared graph (Regime 2). They exist to keep that graph trustworthy as exploration adds to it, and two of them also preserve what exploration found: I_4 requires a hypothesis that crosses experimental models to carry its `ProspectiveFalsifierGate`, the concrete experiment that would test it, and I_6 keeps a claim's supporting and counter-evidence anchors (T^{+} and T^{-}) disjoint, so recorded disagreements between studies stay visible instead of being merged away.

4.1 Six invariants and two-layer package verification

In plain terms: a commit is accepted only if six conditions all hold. Every reference points to something that exists, and the graph matches the curated data exactly (I_1). The edit log replays forward and backward (I_2). Every number, unit, arm, and direction in the claim matches the finding it cites (I_3). A result is carried to another experimental setting only where the context map allows, and a bridging hypothesis names the experiment that would test it (I_4). Nothing is promoted from a figure that has not been digitized or from a source that has not been verified (I_5). Evidence for and against a claim stays separate, and no unchecked text can hide in metadata (I_6).

A CEG package and synthesis claim pass verification iff all six invariants hold:

- ***I_1* (graph integrity, referential closure, and bidirectional corpus binding).** Λ satisfies poset axioms; every context coordinate in $\mathcal{S}$, $\mathcal{A}$, and c_{target} belongs to Λ; every `EvidenceAtom` satisfies Definitions 2–3, with no forbidden Unicode in its identifier, endpoint, unit, arm-identifier, interval/p-value, and provenance strings, balanced brackets in its endpoint name and units, and no confusable homoglyph in its endpoint name, units, and arm identifiers; every anchor tuple in $T^{+} \cup T^{-}$ resolves in $\mathcal{A}$ (`I1_DANGLING_ANCHOR_TUPLE`), and every tuple cited in Σ_{prose} or $\mathcal{P}_{\mathrm{typed}}$ belongs to the record's declared anchor set $T^{+} \cup T^{-}$ (`I1_UNDECLARED_PROSE_TUPLE`, `I1_UNDECLARED_PROPOSITION_TUPLE`); record and proposition identifiers carry no forbidden Unicode (`I1_FORBIDDEN_UNICODE_IN_RECORD_ID`, `I1_FORBIDDEN_UNICODE_IN_PROPOSITION_ID`); and `verify_atoms_bound_to_corpus` confirms **bidirectional parity** between $\mathcal{G}$ and D_m: no orphan study or

outcome exists on either side (`I1_STUDY_WITHOUT_CORPUS_RECORD`, `I1_CORPUS_STUDY_WITHOUT_GRAPH_RECORD`, including zero-outcome corpus entries), and every field recorded in D_m (endpoint, quantity kind, relation, exact value/interval bounds, unit, change direction, focal/comparator arms, statistic kind, sample size, and provenance flags) agrees exactly with $\mathcal{A}$. Provenance flags must be genuine booleans: `EvidenceAtom.validate` refuses any other type (`PROVENANCE_BOOLEAN_MUST_BE_BOOL`, reported as `I1_ATOM_VALIDATION_ERROR`), the corpus comparison refuses a non-boolean flag in D_m (`I1_ATOM_CORPUS_FIELD_MISMATCH`), and the claim check refuses a cited atom whose flag is not a boolean (`I1_INVALID_PROVENANCE_BOOLEAN`). The promotion request flag of a claim must likewise be a boolean (`I1_INVALID_PROMOTION_BOOLEAN`).

- $\boldsymbol{I_2}$ **(bijective ledger replay).** Forward replay of $\mathcal{L}$ from D_0 satisfies all per-step pre/post study digests and old-value guards to produce D_m, and reverse replay of $e_m^{-1}, ..., e_1^{-1}$ from D_m reproduces D_0.
- $\boldsymbol{I_3}$ **(three-tier proposition grounding).**
  - $\boldsymbol{I_3^{\text{tok}}}$ (Unicode-hardened lexical & role-typed token grounding): prose contains no forbidden invisible, combining/enclosing, non-ASCII space, or confusable-script characters; no non-canonical numeral evasions; and every extracted numeric/unit mention is licensed by its scoped atom(s) in a role consistent with its unit and SI prefix case (§4.2).
  - $\boldsymbol{I_3^{\text{clause}}}$ (per-clause contrast, arm-order, inequality, and polarity lint): every depth-0 clause carrying tuple citations grounds its mentions in *that clause's* cited tuple pool (rejecting pooled multi-tuple numeral clauses under strict mode), enforces conservative contrast-connector requirements on multi-arm clauses, and checks arm attribution, negated inequality inversion, and directional polarity cues (§4.2).
  - $\boldsymbol{I_3^{\text{typed}}}$ (exact seven-coordinate proposition homomorphism): every $p \in \mathcal{P}_{\text{typed}}$ matches its target atom on all seven coordinates with duplicate-free arm tuples (Theorem 3), and prom = true requires $\mathcal{P}_{\text{typed}}$ to cover every $t \in T^+$.
- $\boldsymbol{I_4}$ **(partial-order transfer and falsifier-gate completeness).** Positive claims ($\omega \in \{\texttt{SUPPORTS}, \texttt{CANDIDATE_UPGRADE}\}$) require $T^+ \neq \varnothing$ and forbid $\tau_b(c_a, c_{\text{target}}) = \texttt{INADMISSIBLE_TRANSFER}$ on any $t \in T^+$. Direct promotion (prom = true) requires $\omega = \texttt{SUPPORTS}$ and $\tau_b(c_a, c_{\text{target}}) = \texttt{DIRECT_MATCH}$ for every $t \in T^+$. Whenever $U_\cup = \bigcup_{t \in T^+} U(c_a, c_{\text{target}}) \neq \varnothing$ on a `CANDIDATE_UPGRADE` (or whenever a `ProspectiveFalsifierGate` is attached), the gate must name only declared axes of Λ, cover $U_\cup$, specify two distinct non-empty head-to-head arms, and state a non-empty primary readout and a non-empty pass threshold (the gate carries one free-text readout field and one free-text threshold field, `quantitative_pass_threshold`; the verifier checks that both are present and does not parse a unit or a number from them). The three failure codes are `I4_MISSING_PROSPECTIVE_FALSIFIER_GATE` (no gate on a `CANDIDATE_UPGRADE` with $U_\cup \neq \varnothing$), `I4_INCOMPLETE_PROSPECTIVE_FALSIFIER_AXES` (the gate does not cover $U_\cup$), and `I4_INVALID_PROSPECTIVE_FALSIFIER_GATE` (a malformed gate). A non-promoting `SUPPORTS` record over a `BOUNDED_BRIDGE` needs no gate and receives the terminal decision `QUALITATIVE_SCOPING_ONLY`; transfer is classified over T^+ only, so `CONTRADICTS`, `SCOPING_QUALIFICATION`, and `FALSIFIED_GRAVEYARD` records are not rejected for an inadmissible source.
- $\boldsymbol{I_5}$ **(primary-source and digitization gate).** Direct promotion (prom = true) is rejected if any $t \in T^+$ has `requires_figure_digitization = True` or `primary_pdf_verified = False`. Direct promotion is also rejected when no supporting atom holds a numeric quantity, that is, when every cited atom is typed `missing_numeric` (`I5_ILLEGAL_QUANTITATIVE_PROMOTION_OF_MISSING_NUMERIC`).
- $\boldsymbol{I_6}$ **(anchor disjointness, claim attestations, and non-prose surface locks).** $T^+ \cup T^- \neq \varnothing$, $T^+ \cap T^- = \varnothing$, any populated `attestation_sha256` equals the freshly recomputed claim-attestation digest (`compute_claim_attestation_sha256`; a mismatch raises `I6_STALE_CLAIM_ATTESTATION`), any optional `record_envelopes` sequence supplied to `verify_ceg_package` or `compute_ceg_section_digests` corresponds bijectively in order and canonical core-field values to the verified `records` (`verify_envelopes_bound_to_records`, raising `I6_ENVELOPE_RECORD_COUNT_MISMATCH` or `I6_ENVELOPE_RECORD_MISMATCH`), and any serialized claim envelope matches the expected non-prose surface digest passed to `verify_non_prose_surface_lock` across 100% of keys outside `ACTIVE_PROSE_KEYS` (`I6_NON_PROSE_METADATA_TAMPER`). The attestation comparison is evaluated without short-circuiting the semantic I_1–I_5 checks. Within `verify_ceg_package`, both digest comparisons (plus envelope `attestation_sha256` binding) belong to Layer 2 below; the anchor, disjointness, and non-digest envelope-to-record binding conditions are checked in both layers.

Two-layer package verification (`verify_ceg_package`). In plain terms, Layer 1 checks the content itself and rejects structural or semantic tampering even if every stored checksum has been deleted, and Layer 2 adds checksum comparisons that catch changes the content checks cannot see, such as a reworded rationale. To

ensure that cryptographic digests never substitute for structural checking, package verification separates into two explicit layers:

- **Layer 1 (`check_top_digests=False`, structure-only closure):** Verifies graph integrity, every pointer and old-value guard of the ledger, exact equality of the forward- and reverse-replayed corpora with D_m and D_0, bidirectional atom-to-corpus parity, and the non-digest conditions of I_1 and I_3–I_6 on every claim record, **without** consulting any precomputed digest: the per-event study digests, the claim-attestation digests, and the section digests are all ignored. Any structural, catalogue, value, dangling-reference, or semantic tamper fails Layer 1 even if every stored digest is omitted or zeroed.
- **Layer 2 (`check_top_digests=True`, digest locks):** Additionally checks the per-event study `pre`/`post` digests (completing I_2 as defined above), each record's `attestation_sha256`, and, when section digests are supplied, the five locks computed by `compute_ceg_section_digests`: `initial_corpus_sha256` $= H(D_0)$, `settled_corpus_sha256` $= H(D_m)$, `ledger_sha256` $= H(\mathcal{L})$, `atoms_sha256`, and `records_non_prose_sha256`. Layer 2 catches what replay of values cannot see: reordered same-study events and stale attestations through the per-event and attestation digests, and rationale rewording, event renaming, inserted edit-and-revert pairs, and metadata injected outside `prose_sentences` through the section digests.

4.2 Hardened prose lint specification (I_3^{tok} and I_3^{clause})

In plain terms: the lint reads each evidence-anchor sentence as a strict proofreader would. It refuses invisible and look-alike characters (rule 1), reads units with their exact letter case (rule 2), splits the sentence into clauses and requires each clause's numbers to come from the findings that clause cites (rule 3), flips negated

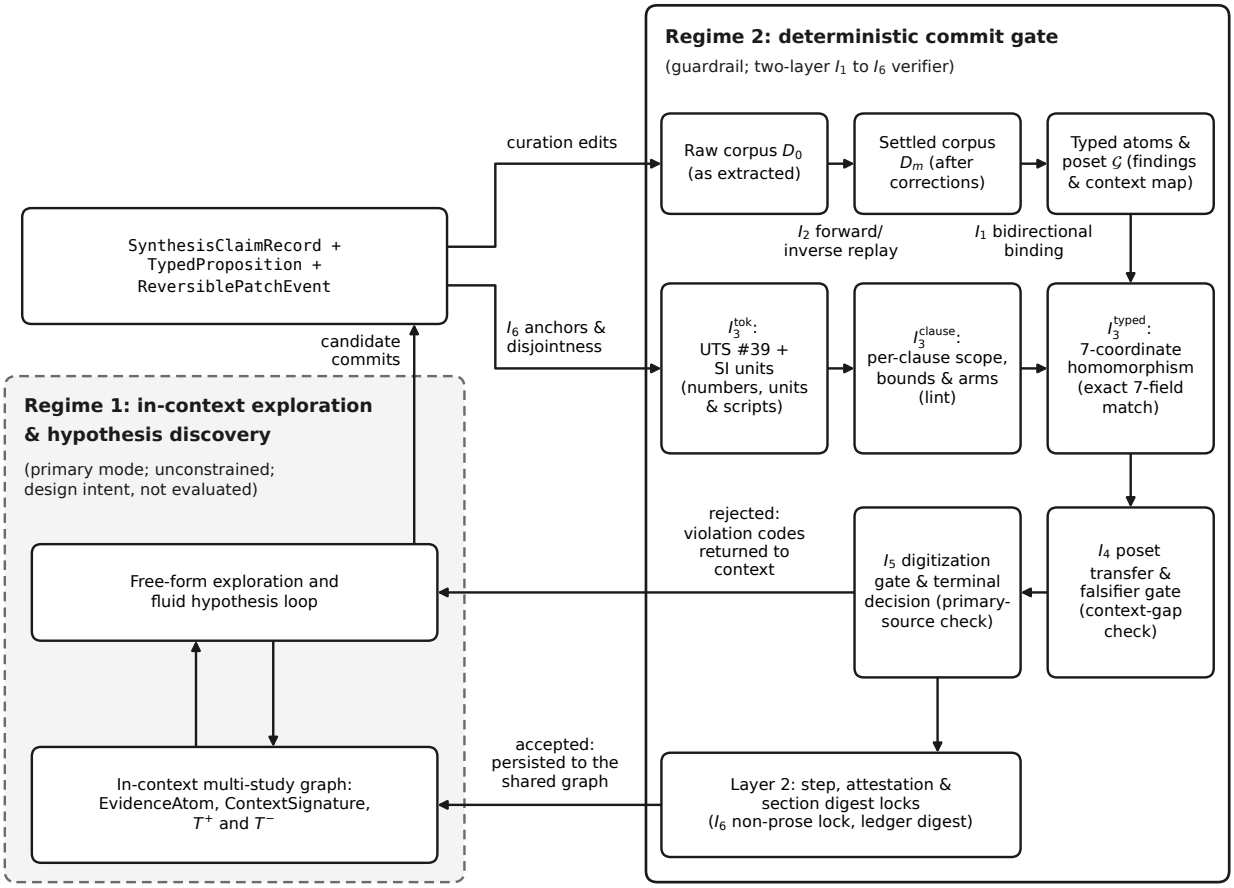


Figure 2. Two-regime workflow and the `V3` commit verifier. Regime 1 (left) is the open-ended exploration and hypothesis-discovery loop over the in-context graph, the primary operating mode; it is a statement of design intent and is neither constrained nor evaluated in this paper. Candidate commits (`SynthesisClaimRecords` with their `TypedPropositions`, and `ReversiblePatchEvent` curation edits) enter Regime 2, the two-layer verifier that acts as its guardrail: Layer 1 checks every structural, replay, lint, typed, and context invariant directly, and Layer 2 adds the per-step, attestation, and section digest locks. Accepted commits persist to the shared graph; rejected commits return structured violation codes to the context for iterative repair.

bounds and requires an explicit comparison word when a clause names both arms of a multi-arm outcome (rule 4), and checks direction words such as *reduced* or *rose* clause by clause (rule 5). The lint is conservative by design: an ambiguous sentence is refused, which is why some true sentences fail (§6.4 and §6.8).

Because prose checks operate on surface strings, every rule and boundary is stated explicitly:

1. **Unicode character and script security (UTS #39 / UTR #36).** `contains_forbidden_unicode` rejects format (`Cf`), control (`Cc` outside `\t`, `\n`, `\r`), surrogate (`Cs`), private-use (`Co`), unassigned (`Cn`), line/paragraph separator (`Zl`, `Zp`), and **all non-ASCII space (Zs)** characters (`U+00A0`, `U+2005`, `U+2009`), any default-ignorable code point, every **enclosing mark (Me, such as U+20DD)**, and any **combining mark (Mn/Mc, such as U+0301) that adjoins an ASCII digit, ASCII letter, decimal separator, sign, whitespace, or a string boundary** (a combining mark between two non-ASCII letters is not flagged). Separately, `contains_confusable_homoglyphs` rejects Cyrillic, Armenian, and Cherokee letters, Greek letters confusable with Latin, mathematical-alphanumeric and fullwidth look-alikes, the Kelvin and Ångström signs, and mixed-script words, both in prose (`I3_TOK_CONFUSABLE_HOMOGLYPH`, for example `PR\u041e-C3` or `r\u043ese`) and in atom endpoint names, units, and arm identifiers at I_1. Scientific symbols remain permitted (±, ×, −, µ/μ), and superscript or subscript digits are accepted inside registered endpoint and unit strings (for example an endpoint `VO₂max` or a unit `mL/min/1.73 m²`); elsewhere in prose they are refused as `I3_TOK_NONCANONICAL_NUMERAL`.
2. **Case-sensitive SI units, `%` right boundaries, and designator guards.** `UNIT_TABLE` distinguishes case-sensitive SI concentration prefixes (`M`, `mM`, `uM`/`µM`, `nM`, `pM`, `fM`) from length prefixes (`m`, `cm`, `mm`, `um`/`µm`, `nm`, `pm`) without case-folding two-letter SI prefix symbols, attaches `%` cleanly before punctuation or parentheses (`94.2%.`, `94.2% (`) without trailing `\b` failure, checks both leading (`PRO-C3 (mg/mL) was 29.4`) and trailing (`concentration was 10 (um)`) parenthesized unit headers, and reads the single-letter units `d`, `h`, `l`, `m`, and `g` only when a space or hyphen separates them from their numeral (`28 d`, `28-d`; a glued `28d` is refused as `I3_TOK_NONCANONICAL_NUMERAL`), suppressing `d` and `h` after a figure/table panel designator (`Fig. 28 d`, `Table 2 h`). Lowercase `l` and `ml` are read as litres and millilitres.
3. **Depth-0 per-clause scoping and contrastive subordinators.** Multi-tuple sentences are split at bracket depth 0 across semicolons, coordinating/contrastive markers (`whereas`, `while`, `but`, and `and` or a clause comma before a citation), **and mid-sentence contrastive subordinators with or without commas (although, even though, though)**, while coordinating citation tails (`(C500:O2) and (C500:O1)`, `; see C500:O2 and C500:O1`) and leading `For C500:O1 and C500:O2,` headers are retained in pooled scope rather than mis-split into single-tuple fragments. Every clause carrying its own tuple citation(s) must ground its numeric/unit tokens in *that clause*'s cited tuple pool—rejecting k-tuple cyclic permutations ($k \geq 3$), one-way cross-clause thefts, and duplicate-token thefts even when every value appears somewhere in the sentence union.
4. **Symbolic/never-negated inequalities and conservative multi-arm binding.** Strict inequality extraction inverts symbolic and verbal negated bounds (`not < 5%` $\rightarrow \geq 5\%$, `not <= 5%` $\rightarrow > 5\%$, `never > 28` $\rightarrow \leq 28$, `never less than 5%` $\rightarrow \geq 5\%$), and conservatively refuses residual bare numeric negation (`not 24 h`) or stacked double negation (`not never < 28 days`): the operator parser returns `AMBIGUOUS_NEGATION`, which matches no atom relation, so the clause fails with `I3_CLAUSE_INEQUALITY_MISMATCH`. On pairwise outcomes, clauses naming an unmodeled study arm alongside a bound arm fail with `I3_CLAUSE_MULTI_ARM_AMBIGUITY`; across multi-arm outcomes, clauses naming both focal and comparator arms without an explicit contrast connector (`vs`, `versus`, `compared with`, `compared to`, `relative to`, `against`, `than`, `in contrast to`, `as opposed to`, `whereas`, `while`, `but`, `even though`, `although`, `though`, `respectively`, `over` introducing an arm reference, or a `from`–`to` trajectory over the two point values) fail with `I3_CLAUSE_NON_CONTRAST_MULTI_ARM` (refusing symmetric *"Arm A and Arm B"* and parenthetical side-notes), and comma-appositive comparative arm swaps (*"the placebo arm had a change, −20.5 mL, compared with treatment"*) are rejected with `I3_CLAUSE_ARM_VALUE_SWAP`.
5. **Clause-local polarity windowing.** Comparator-arm governance resets across `;` or `:` and across `and`, `but`, or `yet` followed directly by a single-token subject before the cue (`_CLAUSE_RESET_RE`), so an absent-reduction clause on a comparator (*"the reduction seen in vehicle slices was absent and PRO-C3 rose to … in treated slices"*) cannot mask an inverted verb on the focal arm (`I3_CLAUSE_DIRECTION_REVERSAL`), trailing polarity clauses after `respectively` lists are checked against all cited atoms, and cues asserted of the comparator arm while denied of the focal arm (*"seen in vehicle slices, not in treated slices"*) fail with `I3_CLAUSE_NEGATED_DIRECTION`.

4.3 Formal results

In plain terms: if the edit log checks out, replaying it forward from the raw extraction gives exactly the curated data and replaying it backward gives exactly the raw extraction; and the checksum-guarded replay fails whenever either endpoint corpus is altered while the other and the edit log are fixed, an edit is dropped or repeated, two adjacent edits to the same study are swapped, or an edit is pointed somewhere else.

Proposition 1 (state-guarded bijective patch replay). *If $(D_0, D_m, \mathcal{L})$ satisfies I_2, then unconditionally for (i), (ii), and (iii)(a),(b),(e), and absent a SHA-256 collision on `RFC 8785` canonical encodings of float-free JSON documents for (iii)(c)–(d): (i) (determinism) forward replay of $\mathcal{L}$ from D_0 yields exactly D_m; (ii) (invertibility) reverse replay of $(e_m^{-1}, ..., e_1^{-1})$ from D_m yields exactly D_0; and (iii) (tamper evidence) digest-guarded replay—`verify_bijective_patch_replay` with its per-step study digests, which is I_2 as defined in §4.1 and what Layer 2 runs—fails for (a) any altered D_0 with $(D_m, \mathcal{L})$ fixed or any altered D_m with $(D_0, \mathcal{L})$ fixed, (b) deletion or duplication of any event e_k, (c) transposition of adjacent events touching the same study, (d) replacement of* ptr_k *by any distinct pointer (including non-canonical index aliases), and (e) any pointer escaping the subtree of study s_k.*

Proof. Canonical `RFC 8785` serialization is injective on float-free JSON trees and deep-copies study subtrees so no mutable container is shared across studies. At step i, forward replay verifies $H(W_{i-1}[s_i]) = H_i^{\mathrm{pre}}$ and $W_{i-1}[\mathrm{ptr}_i] = v_i^{\mathrm{old}}$, writes v_i^{new} at ptr_i to form W_i, and verifies $H(W_i[s_i]) = H_i^{\mathrm{post}}$; exact canonical JSON comparison `_json_equal` verifies $W_m = D_m$ directly without hashing, proving (i). Because ptr_i addresses an existing location holding v_i^{old}, writing v_i^{old} back into W_i restores W_{i-1} and satisfies the swapped guards $(H_i^{\mathrm{post}}, H_i^{\mathrm{pre}})$, proving (ii) by induction from m down to 0. For (iii): (a) with $(D_m, \mathcal{L})$ fixed, any $D_0' \neq D_0$ satisfying all step guards would map by (ii) from its terminal state W_m' back to $D_0' \neq D_0$, forcing $W_m' \neq D_m$ and breaking terminal equality, and symmetrically for $D_m' \neq D_m$ with $(D_0, \mathcal{L})$ fixed; (b) since $v_k^{\mathrm{old}} \neq v_k^{\mathrm{new}}$, omitting or repeating e_k alters $W_k[s_k]$ and breaks either the next guard on s_k or terminal equality with D_m; (c) for adjacent events on s_k, absent a SHA-256 collision $H_k^{\mathrm{pre}} \neq H_k^{\mathrm{post}}$ because $v_k^{\mathrm{old}} \neq v_k^{\mathrm{new}}$, so e_{k+1} encounters $H_k^{\mathrm{pre}} \neq H_{k+1}^{\mathrm{pre}}$; (d) non-canonical pointer tokens fail Definition 5(a) directly, while any distinct canonical pointer either fails the old-value guard, leaves two differing locations in $W_k[s_k]$ (disjoint paths), or would require a finite JSON subtree to equal a proper sub-value of itself (nested paths), breaking H_k^{post} absent a SHA-256 collision (and terminal equality if unoverwritten); and (e) subtree containment is checked directly on ptr_k. ▫

Remark (commuting cross-study edits and Layer-2 digest binding). Adjacent events on *distinct* studies commute and represent the same per-study state transition; replay accepts their transposition unless event digests are hash-chained. Likewise, rewording `rationale`, renaming `event_id`, or inserting an edit followed immediately by its exact inverse preserves the map $D_0 \mapsto D_m$ and is accepted by replay, while altering $H(\mathcal{L})$ and failing Layer-2 digest verification (Table 8). Because per-event digests cover only studies touched by $\mathcal{L}$, an identical modification applied simultaneously to an unpatched study in both D_0 and D_m also preserves the relative transition $D_0 \mapsto D_m$ under standalone replay and is caught in Layer 2 by the whole-corpus endpoint locks (`initial_corpus_sha256`, `settled_corpus_sha256`). Layer 1 verifies pointers, old-value guards, and exact equality of the replayed corpora but does not consult the per-step digests, so part (iii)(c) is not guaranteed there: adjacent events on the *same* study that edit disjoint locations also commute under value-only replay, and their transposition is accepted by Layer 1 and rejected by Layer 2.

In plain terms: (1) provided a result that is valid in one setting stays valid in every setting that it covers, whenever the gate allows a number to be carried directly from one setting to another the transfer is sound, and when the gate does not allow it, it names exactly the axes that fall short; (2) as soon as two settings cannot be ranked against each other, such as a mouse model and human tissue slices, no single evidence score with a cut-off can allow all of the sound transfers and only those.

Theorem 1 (poset transfer soundness and impossibility of a faithful scalar evidence ranking). *(1) Suppose validity of a quantitative parameter within an endpoint family is monotone down $(\mathcal{C}, \preccurlyeq_{\mathcal{C}})$: whenever it holds at $c \in \mathcal{C}$, it holds at every $c' \preccurlyeq_{\mathcal{C}} c$. If I_4 permits direct quantitative promotion from c_s to c_t, then validity at c_s implies validity at c_t; otherwise $U(c_s, c_t)$ is the unique minimal set of axes on which c_s must change to cover c_t. (2) Suppose $\mathcal{C}$ contains at least one incomparable pair $c^{(1)} \parallel_{\mathcal{C}} c^{(2)}$. For any scalar ranking $r : \mathcal{C} \to \mathbb{R}$ and threshold $\theta \in \mathbb{R}$, define $T_{r,\theta}(c, c') \Leftrightarrow r(c) - r(c') \geq \theta$. No rule $T_{r,\theta}$ is simultaneously complete $(\forall c' \preccurlyeq_{\mathcal{C}} c, T_{r,\theta}(c, c'))$ and sound $(\forall c, c', T_{r,\theta}(c, c') \Longrightarrow c' \preccurlyeq_{\mathcal{C}} c)$.*

Proof. (1) I_4 permits direct promotion iff $\tau_b(c_s, c_t) = \texttt{DIRECT_MATCH} \Leftrightarrow U(c_s, c_t) = \varnothing \Leftrightarrow \forall k \in D, c_{t,k} \preccurlyeq_k c_{s,k} \Leftrightarrow c_t \preccurlyeq_{\mathcal{C}} c_s$, whence monotonicity yields validity at c_t. If $k \in U(c_s, c_t)$, any context c' with $c'_k = c_{s,k}$ has $c_{t,k} \not\preccurlyeq_k c'_k$ and fails to cover c_t, whereas setting $c'_k = c_{t,k}$ for $k \in U(c_s, c_t)$ and $c'_k = c_{s,k}$ otherwise satisfies $c_t \preccurlyeq_{\mathcal{C}} c'$. (2) Reflexivity $c^{(1)} \preccurlyeq_{\mathcal{C}} c^{(1)}$ and completeness give $r(c^{(1)}) - r(c^{(1)}) = 0 \geq \theta$. Without loss of generality $r(c^{(1)}) \geq r(c^{(2)})$, so $r(c^{(1)}) - r(c^{(2)}) \geq 0 \geq \theta$, licensing transfer from $c^{(1)}$ to $c^{(2)}$ despite $c^{(2)} \not\preccurlyeq_{\mathcal{C}} c^{(1)}$ [28, 29]. ▫

In plain terms: when one sentence cites two outcomes of the same study that share arms, units, and direction, a checker that looks only at the citations, numbers, units, and arm and direction words cannot tell the true sentence from the one that attaches the right numbers to the wrong outcome, so it must get half of such sentences wrong.

Theorem 2 (surface-verifier indistinguishability under pooled citations). *Let* $\Phi(\text{cl}) = (T_{\text{cl}}, M_{\text{cl}}, \pi_{\text{cl}})$ *be the projection of a clause* cl *onto its resolved citation scope* T_{cl}*, its sequence of extracted numeral–unit mentions* M_{cl}*, and its sequence of arm and polarity cues* π_{cl}*. Call any checker whose verdict on* cl *is a function solely of* $\Phi(\text{cl})$ *and* $\{\mathcal{A}[t] : t \in T_{\text{cl}}\}$ *a* surface verifier*. Let* $o_1 \neq o_2$ *be two outcomes of one study sharing identical arms, units, and polarity, distinct numeric values, and endpoint names contributing identical numerals and cues to* Φ*. Let* s_{ij} *denote the sentence naming endpoint* o_i*, quoting* o_j*'s values in focal-then-comparator order, and citing* (o_1, o_2) *in a pooled scope* $T = \{o_1, o_2\}$*. Then* $\Phi(s_{12}) = \Phi(s_{22})$ *and* $\Phi(s_{21}) = \Phi(s_{11})$*, so every surface verifier errs on exactly two of the four sentences* $\{s_{11}, s_{22}, s_{12}, s_{21}\}$*.*

Proof. Because o_1 and o_2 share arms, units, and direction, s_{11} and s_{22} are true while s_{12} and s_{21} misbind o_j's values to o_i's endpoint. Sentences s_{12} and s_{22} have identical citation scope $\{o_1, o_2\}$, identical numeral–unit sequence $M(o_2)$, and—because their endpoint noun phrases contribute identical numerals and cues—identical π_{cl}. Thus $\Phi(s_{12}) = \Phi(s_{22})$ and symmetrically $\Phi(s_{21}) = \Phi(s_{11})$, forcing any function of $(\Phi, \{\mathcal{A}[o_1], \mathcal{A}[o_2]\})$ to return one verdict on $\{s_{12}, s_{22}\}$ and one verdict on $\{s_{21}, s_{11}\}$, making exactly one error per pair. ▫

In plain terms: on the indistinguishable pooled pairs of Theorem 2, refusing the pooled clause is necessary to block the misbound sentences, so the reference implementation adopts blanket refusal of any numeral-bearing pooled clause as a conservative sufficient policy (which also refuses some true pooled sentences); writing one sentence per outcome avoids the problem, and the reference implementation generates an accepted sentence of that kind for every finding in the benchmark graphs.

Corollary 2.1 (sound pooled policy and single-outcome canonical rendering). *Within the class of surface verifiers, on every indistinguishability class of Theorem 2, rejecting the pooled sentences is necessary to reject all pooled misbindings* s_{12}, s_{21}*, at the cost of refusing the true pooled sentences* s_{11}, s_{22}*; rejecting every numeral-bearing clause whose scope has* $|T_{\text{cl}}| \geq 2$ *(`pooled_numeral_policy="reject"`) is therefore adopted as a conservative sufficient policy across all pooled clauses. Splitting an* m*-outcome pooled claim into* m *single-outcome clauses avoids pooled scope (*$|T_{\text{cl}}| = 1$ *on each clause), and* `render_atom_sentence` *provides a deterministic single-outcome rendering accepted by the strict lint across all 242/242 benchmark and hostile-name atoms (Table 8).*

`render_atom_sentence` relies on lexicon masking rather than an unconditional guarantee over arbitrary strings. The verbatim endpoint phrase and any numeral-bearing unit string are registered in the graph's masking lexicon, so ordinary digits, superscripts, Roman numerals, and direction words inside them are not read as standalone numerals or cues, and the bracket-balance check of I_1 keeps the citation scope resolvable (verified across 242/242 atoms in Table 8). Completability is non-universal when I_1-valid fields collide with prose tokenization or omit rendered operators: `render_atom_sentence` is refused if an outcome local identifier lacks the `O\d+` prose prefix required by `TUPLE_RE`, if a `relative_change` atom carries an inequality relation (`<`, `<=`, `>`, `>=`, which `render_atom_sentence` omits from its template), if an endpoint or unit string itself contains an outcome-tuple identifier, if a unit string contains an unmasked number word (`half`), direction cue (`lower`), or relation operator, or if a short registered endpoint in the graph (such as `>` or `5`) masks an adjacent relation symbol or decimal fraction in the rendered clause. Table 8 verifies single-outcome completability exhaustively on the benchmark graphs rather than claiming it for all strings. Theorem 2 characterizes pooled scopes only; a *single-tuple* clause that cites `P900:O1`, states `P900:O1`'s true numbers, and writes the wrong endpoint noun phrase (`R1` single-tuple templates) remains outside surface checking and motivates grounding decisions on I_3^{typed}.

In plain terms: a structured claim passes only if it matches its finding exactly on seven fields (the endpoint name, the two sets of arms, the two quantities, the statistic, and the sample size), so each finding has exactly one valid structured claim, apart from its identifier, the order in which arms are listed, and letter case or surrounding spaces in the endpoint name, and changing any single field is rejected.

Theorem 3 (unique seven-coordinate exactness of the typed layer). *For $a = \mathcal{A}[t]$, define*

$$\Pi(a) = \big(\ \mathrm{norm}(a.\mathrm{ep}), \mathrm{set}(a.\gamma.A_{\mathrm{focal}}), \mathrm{set}(a.\gamma.A_{\mathrm{comp}}), a.q_{\mathrm{focal}}, a.q_{\mathrm{comp}}, a.\mathrm{stat}, a.n\ \big),$$

where norm *(`normalize_endpoint_name`) trims leading and trailing ASCII spaces and tabs and folds ASCII letters `A–Z` only; internal whitespace is compared as written. A duplicate-free `TypedProposition` p referencing t satisfies I_3^{typed} iff its seven coordinates equal $\Pi(a)$. Consequently: (i) each atom admits a unique valid typed proposition up to proposition ID, arm-tuple permutation, and the ASCII letter case and surrounding ASCII space of* ep*; (ii) every single-coordinate mutation—including pooled endpoint–value swaps, focal/comparator arm-role swaps, comparator-quantity substitution, relative-vs-level scale conflation, statistic changes, and sample-size omissions or float coercions—is rejected; (iii) when* $\mathrm{prom} = \mathrm{true}$*, every $t \in T^+$ is covered by such a proposition; and (iv) verification runs in $O(|A_{\mathrm{focal}}| \log |A_{\mathrm{focal}}| + |A_{\mathrm{comp}}| \log |A_{\mathrm{comp}}| + |\mathrm{ep}| + L_{\mathrm{lit}})$ time per proposition, where L_{lit} is the total character length of the serialized arm identifiers and quantity literals.*

Proof. `verify_typed_proposition_homomorphism` rejects duplicate arm entries inside its two arm checks (a repeated arm raises `I3_TYPED_FOCAL_ARM_MISMATCH` or `I3_TYPED_COMPARATOR_ARM_MISMATCH`), so the remaining comparison on duplicate-free arm tuples is exactly set equality on $\mathrm{set}(A_{\mathrm{focal}})$ and $\mathrm{set}(A_{\mathrm{comp}})$; checks $\mathrm{norm}(p.\mathrm{ep}) = \mathrm{norm}(a.\mathrm{ep})$ without Unicode NFKC folding (so `K` Kelvin sign does not fold to ASCII `k`); compares `QuantitySpec` dataclasses across all nine fields and `sample_size` with strict integer type checking (`type(n) is int`); and enforces $T^+ \subseteq \{p.t : p \in \mathcal{P}_{\mathrm{typed}}\}$ whenever $\mathrm{prom} = \mathrm{true}$. ▫

5. Worked Example: Immune-Evasive Islet Replacement for Type 1 Diabetes

This section follows ten findings from eight peer-reviewed studies through every step: extraction into structured records, classification of each finding against a clinical target setting, logged corrections, and the decision on each candidate claim. The studies, the quoted sentences, the study arms, and the numbers are real and are taken from the production corpus of §6.9. A second, fully synthetic bundle, kept because the benchmark of §§6.1–6.7 is built on it, is described in §5.4.

In plain terms: the point of this section is to show the representation and the checks working on real papers rather than on invented ones. The question it answers is not whether any of these therapies works. It is what a curated record of each finding allows a synthesis to say about a human islet graft, and what it refuses to let a synthesis say.

The `t1d_islet_replacement` vignette (`domain_vignettes.py`) models a team curating evidence on protecting transplanted insulin-producing cells from immune attack, toward the clinical target context

$$\begin{aligned} c_{\mathrm{target}} = (&\texttt{clinical_t1d_islet_graft},\\ &\texttt{immunocompetent_auto_plus_allo_t1d},\\ &\texttt{vascularized_marginal_mass_graft},\\ &\texttt{durable_c_peptide_and_insulin_independence}). \end{aligned}$$

That is: islets transplanted into a person with type 1 diabetes, facing both the returning autoimmune response and the response to donor tissue, engrafted at a marginal cell mass in a vascularized site, and judged by durable C-peptide secretion and insulin independence rather than by a transcript or a surface marker. The four axes of Λ are the model system, the immune challenge, the engraftment microenvironment, and the maturity of the read-out. Across the $6 \times 4 \times 3 \times 3 = 216$ contexts of Λ, evaluating all 46,656 ordered context pairs yields 5,400 `DIRECT_MATCH`, 23,112 `BOUNDED_BRIDGE` and 18,144 `INADMISSIBLE_TRANSFER` classifications. In plain terms, for every ordered pair of settings on this context map, the classifier states whether a finding obtained in the first setting can speak to the second directly, only as a bridged hypothesis, or not at all.

5.1 The ten findings and their transfer classification

Table 1. Findings of the type 1 diabetes worked example and their transfer classification toward c_{target}.

Finding	Setting	Endpoint and contrast	Result as recorded	Transfer class and unresolved axes U
R1_0340:02	Human population genetics, type 2 diabetes [46]	Type 2 diabetes risk, *SLC30A8* loss-of-function carriers vs non-carriers (pairwise)	37% decrease; odds ratio 0.62 (relative_change)	INADMISSIBLE_TRANSFER; engraftment, immune, model; model has no common upper bound
R1_0141:09	Human iPSC-derived beta-like cells in vitro [47]	Number of *INS*- and *SLC30A8*-expressing cells, two mutant-genotype groups (four clones) vs grouped controls (multi_arm_joint, 3 arms)	Fewer cells; no number (missing_numeric)	BOUNDED_BRIDGE; all 4 axes
R1_0116:01	Immunosuppressed clinical islet allograft recipients with type 1 diabetes [48]	Risk factors for failure to reach insulin independence (single_arm)	HLA-A*24, *SLC30A8* CT or TT genotype, and BMI at or above the group median; no effect size (missing_numeric, $n = 40$)	DIRECT_MATCH
R1_0214:04	Stem-cell-derived islets grafted in humanized mice [49]	Guide-RNA enrichment in grafts of mice given human immune cells vs control mice, genome-wide screen (pairwise)	*B2M*, *HLA-A*, *TAP1/2*, *STAT1*, *JAK1/2*, *CXCL10* enriched; no number (missing_numeric, $n = 12$)	BOUNDED_BRIDGE; immune, model, read-out
R1_0214:012	Stem-cell-derived islets in vitro, interferon-gamma [49]	HLA, CXCL10 and inhibitory proteins, *STAT1* knockout vs control (pairwise)	HLA, CXCL10, HLA-E, PD-L1 and SOCS1 all reduced; no number (missing_numeric)	BOUNDED_BRIDGE; all 4 axes
R1_0215:07	Human iPSC-derived beta-like cells in vitro, interferon-alpha [50]	HLA-ABC after *NLRC5* knockdown vs control (pairwise)	Reduced; no number (missing_numeric)	BOUNDED_BRIDGE; all 4 axes
R1_0215:027	Human iPSC-derived islet cells in vitro, interferon-alpha [50]	*BCL2L1* expression, alpha-like vs beta-like cells (pairwise)	0.13 log2 fold change higher in alpha-like cells (relative_change)	BOUNDED_BRIDGE; all 4 axes
R1_0098:08	Human ESC-derived cells, endocrine cells matured in mice, assayed in vitro [51]	CD55 expression, endocrine cells vs pancreatic endoderm (pairwise)	Clearly expressed on endocrine cells, lower on endoderm; no number (missing_numeric)	BOUNDED_BRIDGE; all 4 axes
R1_0202:07	INS-1 rat beta-cell line [52]	Glucose-stimulated insulin secretion, *CD59* knockout vs control (pairwise)	Strongly inhibited; no number (missing_numeric, $n = 5$)	INADMISSIBLE_TRANSFER; all 4 axes; model is below the floor
R1_0376:015	Stem-cell-derived beta cells in vitro, hypoxia [53]	*TXNIP* expression, hypoxia vs normoxia (pairwise)	Upregulated; no number (missing_numeric)	BOUNDED_BRIDGE; all 4 axes

In plain terms: only one of the ten findings is classified as a direct match to the declared target coordinates, and it comes from an immunosuppressed clinical allograft cohort [48] whose record, which rests on the abstract alone, lists the factors that predicted failure to reach insulin independence without an effect size for them. Six findings come from human stem-cell-derived cells assayed in culture and one from a graft in a mouse given a human immune system; each of those can support a bridged hypothesis about a human graft, never a transferred number. The two findings the classifier refuses are instructive. A large human genetic study of type 2 diabetes risk sits on a branch of the model-system axis that shares no common upper bound with a transplanted islet graft, so its odds ratio cannot be carried over. A result in an immortalized rat beta-cell line sits below the declared bridge floor, so it cannot be bridged either. A finding that cannot be carried over as support is still recorded, and both of these are cited as counter-evidence in §5.3.

What is and is not verified about these findings. Every finding carries the verbatim sentence from which it was extracted. For this paper, all ten quoted sentences were located in the article text held by the National Center for Biotechnology Information: nine in PubMed Central full text, and one in a PubMed abstract, for the one study without a PubMed Central record (`t1d/check_t1d_vignette.py --quotes`). That check locates the quoted sentence only. It does not confirm the study arms, the direction, or the sample size, and it reads no PDF, so `primary_pdf_verified` is `false` on all ten findings and I_5 blocks every exact quantitative promotion in this bundle. The position of each finding on the four axes, which decides its transfer class, is the curator's reading of the study design and not a field of the corpus. In particular, the forty recipients in R1_0116:01 received intraportal allografts under systemic immunosuppression [48]; assigning both that cohort and

c_{target} the coordinate `immunocompetent_auto_plus_allo_t1d` abstracts the presence of recipient autoimmune and allogeneic reactivity rather than absence of immunosuppression, whereas targeting an immunosuppression-free immune-evasive graft would leave treatment regimen as an unresolved bridge distinction. For the type 2 diabetes genetics study only the model-system coordinate carries meaning: a population cohort has no natural value on the other three axes, so those coordinates are placeholders, and its transfer class does not depend on them. The reading of the axes is the part of this example most open to disagreement, which is why the Discussion describes the context poset as a modeling commitment to be reviewed together with the graph.

5.2 Curation replay from D_0 to D_3

The constructed starting bundle D_0 (SHA-256 prefix `5c38c177c0b7`) contains 8 studies, 20 arm variables and 10 evidence atoms. Three reversible `JSONPatch` events reach the settled corpus D_3 (SHA-256 prefix `fd1e9336453a`): `T1D-PATCH-001` retypes the 37% decrease in type 2 diabetes risk in `R1_0340:O2` from `"point_level"` to `"relative_change"`, because a 37% decrease is a relative change and not a measured level; `T1D-PATCH-002` restores the dropped second mutant arm `VAR:R1_0141:A8_iPSC_Lys34_hom` in the three-arm comparison `R1_0141:O9`; and `T1D-PATCH-003` sets `requires_figure_digitization = true` on `R1_0215:O7`, which is read from a figure panel. Forward and inverse replay verify with zero errors, and all ten atoms satisfy I_1 binding against D_3.

These three defects are constructed for this example. D_0 was built by perturbing the settled records so that the replay has something to repair; the corrections are not errors found in the OpenIslet corpus, whose own 106-event curation ledger is replayed separately in §6.9. The settled values of all three, like the numbers, sample sizes and arm identifiers of Table 1, are the corpus values.

5.3 Synthesis decisions and rejected probes

Five synthesis records verify against the settled bundle, with two terminal decisions between them. `T1D-CLIN-DIRECT-01` → `QUALITATIVE_SCOPING_ONLY` records the one finding classified at the declared target coordinates, the immunosuppressed clinical cohort in which recipient *SLC30A8* genotype, HLA-A*24 positivity and body-mass index were independently associated with failure to reach insulin independence; because that record, which rests on the abstract alone, holds no effect size, it commits as a scoping note and asks for no number. The remaining four are cross-study hypotheses about a human islet graft, each built from evidence obtained outside that setting and each therefore `HOLD_FOR_PROSPECTIVE_FALSIFIER` behind a gate that names its unresolved axes, a head-to-head comparison, a read-out (with units in these four gates) and a pass threshold (numeric in these four gates, although the verifier requires only non-empty strings).

- `T1D-CEG-H1-SLC30A8-INVERSION` → `HOLD_FOR_PROSPECTIVE_FALSIFIER` (polarity `SCOPING_QUALIFICATION`) qualifies the protection against type 2 diabetes conferred by losing one copy of *SLC30A8*. For an islet replacement setting, two records motivate caution without establishing that donor-cell *SLC30A8* knockout causes graft loss: fewer *INS*- and *SLC30A8*-expressing cells form in vitro from stem-cell lines carrying premature-stop genotypes [47], and in the immunosuppressed clinical cohort [48] the recipient's common *SLC30A8* CT or TT genotype was independently associated with failure to achieve insulin independence (a recipient-genotype scoping signal on insulin independence rather than direct evidence about an engineered donor graft). Its gate (`GATE-H1-SLC30A8-MARGINAL-MASS`) asks for beta-like cell yield and twelve-week stimulated C-peptide from edited and unedited stem-cell-derived islets.
- `T1D-CEG-H2-NLRC5-UNCOUPLING` → `HOLD_FOR_PROSPECTIVE_FALSIFIER` (polarity `CANDIDATE_UPGRADE`) proposes testing partial *NLRC5* attenuation against *STAT1* deletion: *NLRC5* knockdown lowers interferon-induced HLA-ABC (`R1_0215:O7`), whereas *STAT1* knockout reduces HLA class I and CXCL10 together with the inhibitory ligands HLA-E, PD-L1, and SOCS1 (`R1_0214:O12`, carried as counter-evidence). Because the same *NLRC5* study [50] also observed reduced *HLA-E* and 24-hour *PD-L1* expression after *NLRC5* knockdown in islet and beta-cell models (additional corpus outcomes outside this ten-finding subset), selective preservation of inhibitory ligands under partial attenuation is an unresolved empirical challenge rather than an established separation; its gate (`GATE-H2-NLRC5-VS-STAT1KO`) requires head-to-head suppression of HLA class I alongside retained HLA-E and PD-L1 surface expression and natural-killer-cell survival.
- `T1D-CEG-H4-CD55-ARMOR-PRESERVE-CD59` → `HOLD_FOR_PROSPECTIVE_FALSIFIER` (polarity `CANDIDATE_UPGRADE`) proposes protecting cells with ectopic CD55, a complement regulator that is lower at the pancreatic endoderm stage than on endocrine cells, while leaving endogenous CD59 untouched, because knocking CD59 out

strongly inhibits glucose-stimulated insulin secretion in a beta-cell line. Its gate (GATE-H4-CD55-ECTOPIC-CD59-UNTOUCHED) pairs survival in human serum complement with a dynamic perifusion secretion index.

- T1D-CEG-H5-TXNIP-HYPOXIA-UNCOUPLING → HOLD_FOR_PROSPECTIVE_FALSIFIER (polarity CANDIDATE_UPGRADE) proposes editing the *TXNIP* promoter in stem-cell-derived beta cells, which upregulate *TXNIP* under the hypoxia a fresh graft meets, and reads the higher *BCL2L1* expression of alpha-like cells as a sign that the two cell types differ in stress reserve. The *BCL2L1* finding is a 0.13 log2 fold difference measured under interferon-alpha and not under hypoxia, which is the kind of cross-condition step the gate exists to hold. Its gate (GATE-H5-TXNIP-CIS-EDIT) measures the ratio of insulin- to glucagon-positive cells fourteen days after a hypoxic episode.

In plain terms: four cross-study hypotheses are written as candidate claims, and the gate holds all four. Each is a statement about a human islet graft built from evidence obtained outside that setting, so each must name the head-to-head experiment that would test it, with a read-out and a pass threshold, before it can be written as anything stronger than a candidate. The fifth record is the one finding classified at the declared target coordinates; because its record holds no effect size, it commits as a scoping note and not as a number. Whether any of these four hypotheses is correct is not assessed here, and the experimental comparators and thresholds in their gates—for example whether H4 also includes an unedited wild-type control alongside CD59-edited cells or whether H5 pairs its cell-type ratio with absolute beta-cell survival and secretion—are proposals requiring domain review, because I_4 checks structural completeness of a falsifier gate rather than experimental-design sufficiency.

Four defective claims are submitted against the same bundle, and all four adversarial probes fail with their exact expected invariant codes. Promoting the type 2 diabetes odds ratio onto the islet-graft target is rejected as I4_INADMISSIBLE_CONTEXT_TRANSFER, and two further codes are raised alongside it. Writing an upgrade hypothesis without a prospective falsifier gate is rejected as I4_MISSING_PROSPECTIVE_FALSIFIER_GATE. Reading the odds ratio of 0.62 as a 62% decrease, when the source reports a 37% decrease, is rejected as I3_TOK_UNGROUNDED_VALUE. Requesting a direct quantitative promotion from a finding whose source has not been certified is rejected as I5_ILLEGAL_QUANTITATIVE_PROMOTION_WITHOUT_PRIMARY_OR_DIGITIZED_SOURCE. Because that finding also holds no number, I5_ILLEGAL_QUANTITATIVE_PROMOTION_OF_MISSING_NUMERIC is raised alongside it.

In plain terms: the third probe catches a subtle numeric misreading. The primary genetics study [46] reports both a 37% decrease in type 2 diabetes risk and an additive-model odds ratio of 0.62—two distinct, non-equivalent summary measures, because an odds ratio is not a risk ratio and $1 - 0.62 = 0.38 \neq 0.37$ (the vignette atom records the 37% relative_change quantity alongside the corpus entry's odds_ratio statistic tag). Reading the 0.62 odds ratio as a 62% decrease invents an unsupported percentage in a sentence that still cites the right study.

5.4 The synthetic NX-410 bundle used by the benchmark

The benchmark of §§6.1–6.7 is built on a second, fully synthetic bundle, kept unchanged so that its cases remain comparable with earlier releases. The drug_development vignette models a translational team curating preclinical and early clinical readouts for an invented oral antifibrotic candidate, NX-410, toward a Phase IIb clinical target context

$$c_{\text{target}} = (\texttt{human_clinical_cohort}, \texttt{established_chronic_fibrosis}, \texttt{oral_steady_state_bid}, \texttt{spirometric_clinical_endpoint}).$$

The drug, the studies, and the numbers are invented for illustration. Across the $6 \times 3 \times 3 \times 3 = 162$ contexts of its lattice, evaluating all 26,244 ordered context pairs yields 3,240 DIRECT_MATCH, 6,462 BOUNDED_BRIDGE and 16,542 INADMISSIBLE_TRANSFER classifications. Its raw corpus D_0 (SHA-256 prefix a12490cea51f) contains 5 studies, 11 arm variables and 9 evidence atoms, and three reversible JSONPatch events reach settled corpus D_3 (SHA-256 prefix 93b4de08b9a8): PATCH-001 retypes S101:O1's 72.0% COL1A1 mRNA reduction from "point_level" to "relative_change"; PATCH-002 restores the dropped active comparator VAR:S102:Pirfenidone_100mgkg in three-arm murine study S102:O1; and PATCH-003 sets requires_figure_digitization = true on Kaplan–Meier curve readout S102:O2 so I_5 blocks exact quantitative promotion.

Table 1b. Evidence atoms of the synthetic NX-410 bundle and transfer classification toward its Phase IIb target context.

Outcome	Model and exposure	Endpoint and contrast	Focal versus comparator	Transfer class and unresolved axes U
S101:O1	Immortalized 2D co-culture, 10 μM bath	COL1A1 transcript (pairwise)	72.0% decrease (relative_change)	INADMISSIBLE_TRANSFER; all 4 axes unresolved, 3 below floor
S102:O1	Murine bleomycin, exposure-matched	Lung hydroxyproline (multi_arm_joint, 3 arms)	142.5 vs $[176.0, 214.0]$ μg/lung	BOUNDED_BRIDGE; all 4 axes
S102:O2	Murine bleomycin, exposure-matched	Day-28 survival (pairwise, undigitized curve)	>28 days (censored_observation)	BOUNDED_BRIDGE; all 4 axes; held by I_5
S103:O1	Human IPF lung slices, exposure-matched	Secreted PRO-C3 (pairwise)	29.4 vs 48.6 ng/mL ($n = 18$)	BOUNDED_BRIDGE; model, exposure, read-out
S103:O2	Human IPF lung slices, exposure-matched	Slice ATP viability (pairwise)	96.2% vs 95.8% ($n = 18$)	BOUNDED_BRIDGE; model, exposure, read-out
S104:O1	Phase IIa, 40 mg BID	12-week FVC change (pairwise)	-28.5 vs -84.0 mL ($n = 64$)	DIRECT_MATCH
S104:O2	Phase IIa, 40 mg BID	12-week supine vital capacity (pairwise)	-31.0 vs -79.5 mL ($n = 64$)	DIRECT_MATCH
S104:O3	Phase IIa, 40 mg BID	Grade ≥ 2 gastrointestinal events (pairwise)	18.8% vs 9.4% ($n = 64$)	DIRECT_MATCH
S105:O1	Phase Ib, 120 mg BID	ALT $> 3 \times$ ULN (pairwise)	25.0% vs 0% ($n = 36$)	BOUNDED_BRIDGE; read-out

Because its numbers are invented, this bundle can exercise decisions that the real one cannot: three valid synthesis records verify with three distinct terminal decisions (SYN-CLIN-01 → PROMOTE_QUANTITATIVE_PARAMETER, promoting the Phase IIa lung-function result directly and recording the gastrointestinal events as counter-evidence; SYN-BRIDGE-02 → HOLD_FOR_PROSPECTIVE_FALSIFIER, combining complementary murine S102:O1 and human-slice S103:O1/S103:O2 evidence behind a four-axis falsifier gate; and SYN-SCOPE-03 → QUALITATIVE_SCOPING_ONLY, recording that the higher dose is excluded because of liver-enzyme elevations). All five adversarial vignette probes fail with their exact expected invariant codes (I4_INADMISSIBLE_CONTEXT_TRANSFER, I3_CLAUSE_CROSS_OUTCOME_SWAP, I3_CLAUSE_AMBIGUOUS_POOLED_CITATION + I3_TYPED_*, I5_ILLEGAL_QUANTITATIVE_PROMOTION_WITHOUT_PRIMARY_OR_DIGITIZED_SOURCE, and I6_SUPPORT_COUNTEREVIDENCE_OVERLAP): a cell-culture result offered as support for a patient-level hypothesis, two sentences that attach one outcome's numbers to another outcome (one of them under a citation covering both outcomes), a promotion that relies on a survival curve that has not been digitized, and a claim that lists the same finding as both support and counter-evidence. A companion physiological vignette (sport_science, 100 ordered context pairs) passes all checks identically. The type 1 diabetes bundle of this section is not part of the benchmark or of the exhaustive self-tests of Table 8, which run on the synthetic bundles; it is checked by its own script (Code and Data Availability).

6. Empirical Evaluation

Scope of the evaluation. Every experiment in this section exercises the deterministic commit gate (Regime 2 of §1): candidate claim records, typed propositions, evidence-anchor sentences, and curation ledgers are submitted to the verifier and its verdicts are counted. No experiment measures upstream exploration (Regime 1), and none estimates how often exploration yields a correct or novel hypothesis. Section 6.9 is a case study on a production corpus: it reports size, structure, and the replay of a correction log, and it does not evaluate hypothesis quality either.

6.1 Benchmark architecture, ablation regimes, and dual-mode protocol

The synthetic benchmark (ceg_synthetic_benchmark.py) comprises 1,931 cases—1,521 adversarial and 410 valid—instantiated from 199 distinct templates across five suites:

1. **Core suite** (800 cases, 37 templates): ten adversarial strata F1–F10 (60 each) and four valid strata P1–P4 (50 each).
2. **Stress suite** (800 cases, 59 templates): ten harder adversarial strata F11–F20 (60 each) and four valid strata P5–P8 (50 each), where P5 (pooled citations) and P7 (derived differences, unsigned magnitudes of negative changes, dispersion ± terms, unmodeled trial-phase numerals, and cross-noun negation scope) measure the conservative lint's refusal boundary.

3. **Residual suite** (240 cases, 12 templates): four open-class adversarial prose strata R1–R4 (60 each) designed to document surface-lint blind spots (misnamed endpoint nouns with true numbers, unitless sample-size swaps, qualitative overstatements, and unlisted implicit negations).
4. **Red-team suite** (43 sentences, 43 templates): an independent adversarial pass, namely variations across T5 and T11–T18 authored by an independent reviewer against frozen release v2.0.
5. **Gauntlet suite** (48 cases, 48 templates: 38 adversarial across G1–G5 and 10 valid controls in G0): a domain-agnostic boundary stress suite, with probes targeting 3-tuple cyclic permutations, one-way/duplicate cross-clause value thefts, although/though subordinators, comma-appositive swaps, Unicode Mn/Me/Zs and UTS #39 homoglyphs, SI prefix case collisions (uM vs um), panel designator collisions (Fig. 28 d), symbolic/never-negated inequalities, conservative multi-arm binding, and colon-bearing PMID:/DOI: identifiers.

Five verification regimes (two baselines and three configurations of the verifier) evaluate every case: V0 (citation resolution only), V1 (V0 + study-wide ASCII numeral containment), V2 (prose lint $I_1 + I_3^{\text{tok}} + I_3^{\text{clause}}$ with pooled_numeral_policy="permit"), V2s (prose lint with pooled_numeral_policy="reject"), and V3 (full CEG verifier: I_1–I_6, strict lint, I_3^{typed}, I_4, I_5, and I_6). In plain terms, the regimes are five checkers of increasing scope, from one that asks only whether the cited finding exists (V0) to the full verifier (V3), so comparing them shows what each added layer catches. They are not strictly nested: V2 does not include the study-wide numeral test of V1, so V1 refuses some valid sentences that V2 accepts (Table 4b, P6). In the tables that follow, a higher count is better in every column: adversarial cases should be rejected, and valid cases should be accepted. A valid case that is refused is a cost of conservative linting and is not an error that reaches the graph.

Dual-mode ("Mode B self-attesting adversary") protocol. In plain terms, Mode A tests a naive edit, which a stale checksum alone would catch, whereas Mode B tests an edit that recomputes fresh checksums to disguise itself, so a Mode B rejection shows that the verifier caught the underlying scientific error rather than relying on a hash. When synthesis records carry precomputed attestation digests (attestation_sha256) and metadata hashes, mutating prose or fields without recomputing attestations (**Mode A: raw tamper**) could appear to reject an attack merely because the attestation hash goes stale. The benchmark therefore evaluates all 1,161 record-level adversarial cases (the adversarial cases of the core, stress, red-team and gauntlet suites whose defect resides inside the claim record rather than in corpus or graph corruption; the residual suite of §6.5, in which 220 of 240 cases pass as predicted, is not part of this count) under **both Mode A and Mode B**, where Mode B re-runs populate_claim_attestation after tampering and asserts that every rejection carries a genuine semantic invariant code (I_1–I_6 disjointness) rather than a stale-attestation error.

Development disclosure. Earlier iterations of the reference implementation serve as pre-hardening baselines from benchmark development. The first release scored 600/600 and 200/200 on the core suite written alongside it, but rejected only 40/600 stress adversarial cases and accepted 107/200 stress valid cases. Frozen release v2.0 closed most stress families but rejected only 1/43 on the independent red-team pass; frozen release v2.1 rejected 580/600 on stress adversarial, 36/43 on red-team, and 20/38 on the new gauntlet adversarial suite (while falsely rejecting 2 of 10 gauntlet valid controls: 8/10 accepted). All synthetic suites are therefore in-sample regression benchmarks for v3.1, whereas §6.8 evaluates primary-source clinical trial abstracts from PubMed.

6.2 Aggregate synthetic results

Table 2. Synthetic benchmark cases rejected (adversarial) and accepted (valid) by suite and regime.

Regime	Core: adv. rejected	Core: valid accepted	Stress: adv. rejected	Stress: valid accepted	Residual: adv. rejected	Red-team: adv. rejected	Gauntlet: adv. rejected	Gauntlet: valid accepted
V0 citation presence	0/600 (0.0%)	200/200 (100.0%)	0/600 (0.0%)	200/200 (100.0%)	0/240 (0.0%)	0/43 (0.0%)	0/38 (0.0%)	10/10 (100.0%)
V1 study-wide bag of numbers	90/600 (15.0%)	200/200 (100.0%)	10/600 (1.7%)	153/200 (76.5%)	0/240 (0.0%)	1/43 (2.3%)	7/38 (18.4%)	10/10 (100.0%)
V2 prose lint, pooled numerals permitted	300/600 (50.0%)	200/200 (100.0%)	420/600 (70.0%)	150/200 (75.0%)	0/240 (0.0%)	40/43 (93.0%)	38/38 (100.0%)	10/10 (100.0%)
V2s prose lint, pooled numerals rejected	360/600 (60.0%)	200/200 (100.0%)	480/600 (80.0%)	100/200 (50.0%)	20/240 (8.3%)	43/43 (100.0%)	38/38 (100.0%)	10/10 (100.0%)
V3 full CEG	600/600 (100.0%)	200/200 (100.0%)	600/600 (100.0%)	100/200 (50.0%)	20/240 (8.3%)	43/43 (100.0%)	38/38 (100.0%)	10/10 (100.0%)

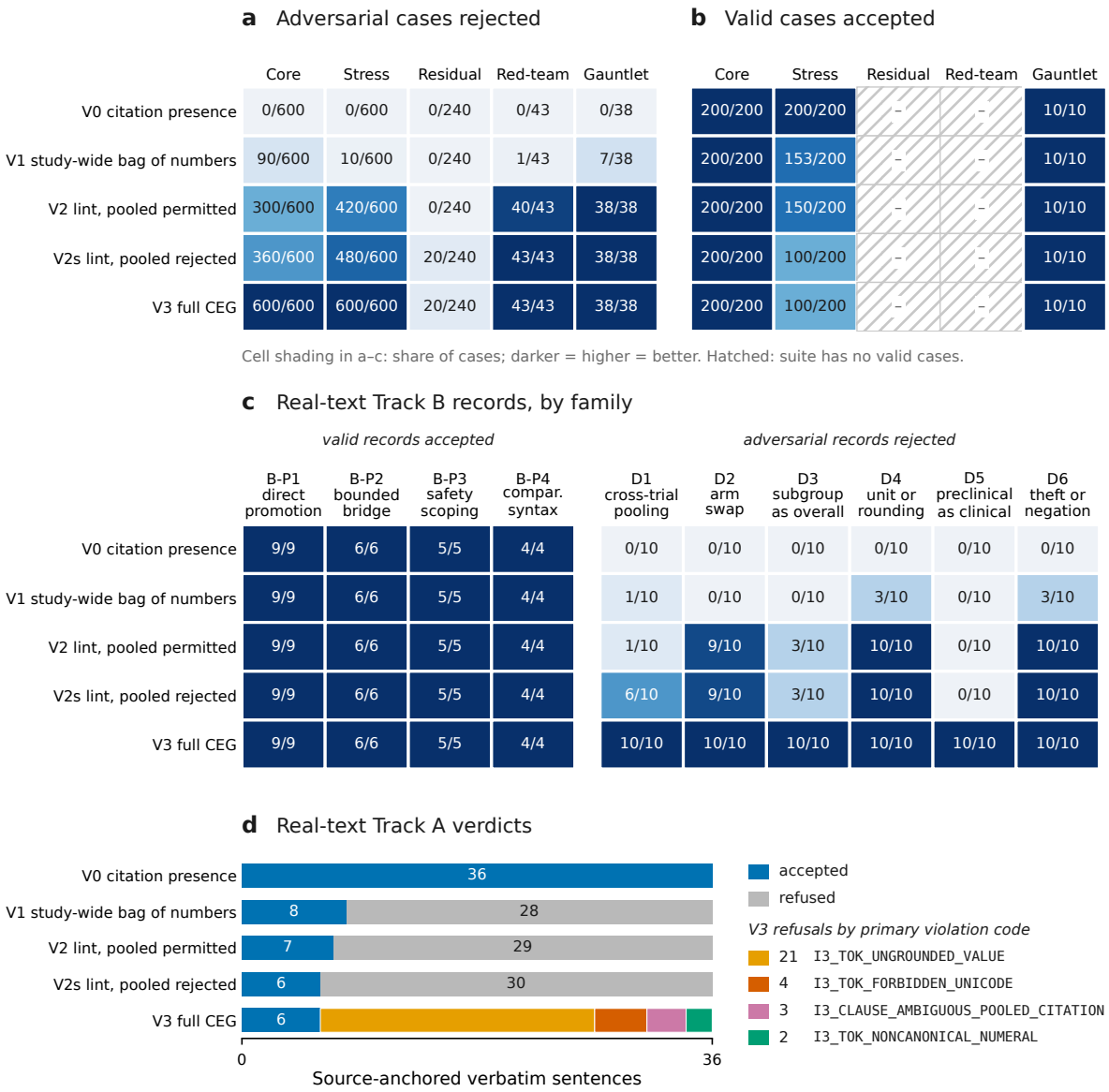


Figure 3. Summary of the empirical evaluation of the commit gate, plotted from `benchmark_results.json` *and* `real_text_results.json` *only. (a) Adversarial cases rejected and (b) valid cases accepted, by synthetic suite (core, stress, residual, red-team, gauntlet) and ablation regime (*`V0`*,* `V1`*,* `V2`*,* `V2s`*,* `V3`*); the counts are those of Table 2. The red-team suite is an independent adversarial pass and the gauntlet suite is a domain-agnostic boundary stress suite (§6.1). (c) Real-text Track B records by family (*`B-P1`*–*`B-P4` *accepted,* `D1`*–*`D6` *rejected) and regime; the counts are those of Table 9b. (d) Real-text Track A verdicts on source-anchored verbatim sentences by regime, with the* `V3` *refusals divided by primary violation code; the counts are those of Table 9a and §6.8.*

Figure 3 summarizes Table 2 together with the real-text results of §6.8 (Tables 9a and 9b).

6.3 Core suite (F1–F10 and P1–P4)

Table 3. Core suite, adversarial families: cases rejected of 60.

Stratum	Mechanism	Templates	V0	V1	V2	V2s	V3
F1	Fabricated numeral absent from the cited study	1	0	60	60	60	60
F2	Sibling outcome's numeral under the cited outcome	1	0	0	60	60	60
F3	Word numerals, vulgar fractions, invisible-character splits	12	0	30	60	60	60
F4	Values swapped across clauses or a `respectively` list	7	0	0	60	60	60
F5	Arm order, `from X to Y` order or verb inverted	3	0	0	60	60	60
F6	Pooled citation misbinds in prose and in the proposition	1	0	0	0	60	60
F7	Relative change typed as level; comparator arm dropped	2	0	0	0	0	60
F8	Promotion or upgrade across an unresolved or inadmissible gap	2	0	0	0	0	60
F9	Promotion from an undigitized figure or unverified source	2	0	0	0	0	60
F10	Support/counter overlap; settled corpus altered after curation	2	0	0	0	0	60

Table 3b. Core suite, valid strata: cases accepted of 50.

Stratum	Mechanism	Templates	V0	V1	V2	V2s	V3
P1	Matched-context promotion with typed proposition	1	50	50	50	50	50
P2	Bridged claim behind a complete falsifier gate	1	50	50	50	50	50
P3	Two-tuple `respectively` sentence	1	50	50	50	50	50
P4	Counter-evidence safety record	1	50	50	50	50	50

Study-wide numeric pooling (V1) rejects 90/600 core adversarial cases (all 60 fabricated literals in F1 plus 30 invisible-split cases in F3 whose digit fragments happen not to occur in the study pool). Permissive prose linting (V2) rejects 300/600 (F1–F5), strict pooled linting (V2s) adds F6 to reach 360/600, and full CEG verification (V3) rejects 600/600 by enforcing typed propositions, contrast non-truncation, partial-order transfer gating, primary-source digitization gates, and ledger replay across F7–F10.

6.4 Stress suite (F11–F20 and P5–P8)

Table 4. Stress suite, adversarial families: cases rejected of 60.

Stratum	Mechanism	Templates	V0	V1	V2	V2s	V3
F11	Pooled prose misbinds while every proposition is true or absent	3	0	0	0	60	60
F12	Promoted tuple uncovered; sample size omitted; comparator misbound	3	0	0	0	0	60
F13	Sibling numeral in a clause fragment that cites nothing	3	0	0	60	60	60
F14	`respectively` list shifted, cross-paired or verb-inverted	3	0	0	60	60	60
F15	True numeral with a unit of another scale	4	0	0	60	60	60
F16	Sample size, name digit or p-value given the outcome's unit	3	0	0	60	60	60
F17	Glued, teen-word, adverbial, superscript or leading-dot numerals	6	0	10	60	60	60
F18	Inversion by irregular verb, noun, negated participle, comparative	4	0	0	60	60	60
F19	Pointer aliases, mis-attributed or no-op events, atom drift	5	0	0	0	0	60
F20	Inverted direction beside a cue of the true direction	3	0	0	60	60	60

Table 4b. Stress suite, valid strata: cases accepted of 50.

Stratum	Mechanism	Templates	V0	V1	V2	V2s	V3
P5	True sentence under a pooled same-unit citation	2	50	50	50	0	0
P6	Natural forms the hardened lint accepts	7	50	43	50	50	50
P7	Magnitude, derived, contextual or dispersion numerals; negation scope	5	50	10	0	0	0
P8	Canonical rendering of an atom	8	50	50	50	50	50

On the stress adversarial suite, V2 rejects 420/600, V2s rejects 480/600, and V3 rejects 600/600 (improving from 580/600 in release `v2.1` by resetting comparator-arm governance across clause boundaries in F20). On stress valid controls, V2 accepts 150/200 and V3 accepts 100/200: P5 (50 true pooled sentences) is refused by Corollary 2.1's `reject` policy—the exact price of rejecting F11—and P7 (50 sentences containing derived differences, unsigned magnitudes of negative changes, ± standard deviations, unmodeled trial-phase numerals, or cross-noun negation scope) is refused because single-atom grounding does not compute cross-value arithmetic or full syntactic dependency trees.

6.5 Residual open-class suite (R1–R4)

Table 5. Residual suite, adversarial families: cases rejected of 60.

Stratum	Mechanism	Templates	V0	V1	V2	V2s	V3
R1	Right tuple and numbers, wrong endpoint name	3	0	0	0	20	20
R2	Sample size, name digit or p-value as the value, no unit	3	0	0	0	0	0
R3	Overstatement or inversion with no numeral or listed cue	3	0	0	0	0	0
R4	True direction named, then denied without a listed negation	3	0	0	0	0	0

V2s and V3 reject 20/240 residual cases: specifically, the 20 instances of R1's pooled-header template (*"For C901:O1 and C901:O2, mean forced vital capacity change was …"*), which release v2.1 mis-split across `and` into a single-tuple scope and which v3.1 properly retains as a pooled multi-tuple header. The remaining 220/240 cases—single-tuple endpoint misnamings (the two single-tuple R1 templates), unitless role swaps (R2), qualitative overstatements (R3), and implicit denials outside closed negation lists (R4)—pass as expected, confirming that prose linting is not a semantic parser and that downstream consumers must rely on I_3^{typed}.

6.6 Independent red-team and gauntlet suites (T5–T18 and G0–G5)

Table 6. Red-team suite across frozen release v2.0, frozen release v2.1 (V2), and current release v3.1 (V3).

Stratum	Variation of	Mechanism	Cases	Rejected by v2.0	Rejected by v2.1 (V2)	Rejected by v3.1 (V3)
T5	F5	Arm attribution, unlabeled value order, `from X to Y` variants	6	0	6	6
T11	F11	Pooled misbinding cited in brackets, separate parentheses, or a trailing clause	3	0	1	3
T14	F14	`respectively` lists with reversed pairs, arms, trajectories or a trailing verb	5	0	4	5
T15	F15	Untabulated, detached or parenthesized units of another scale	8	0	7	8
T16	F16	Sample size or p-value as the value, in word form or with a detached unit	7	1	6	7
T17	F17	Fused `-fold` words, ordinal fractions, multipliers, suffixes, glued integers	8	0	7	8
T18	F18	Comparatives on the other arm, distant negation, `non-` prefix	6	0	5	6
All			**43**	**1**	**36**	**43**

Frozen release v2.0 rejected 1/43 red-team sentences; release v2.1 rejected 36/43; and release v3.1 rejects 43/43 by closing coordinated citation tails (`(C500:O2) and (C500:O1)`, `; see C500:O2 and C500:O1`), leading and trailing parenthesized unit headers (`PRO-C3 (mg/mL)`, `viability (%)`), unregistered alphanumeric-glued integers (`ELISA51`), post-`respectively` inverted verbs, and comparator-reassigned polarity cues. Because v3.1 was revised with access to these 43 sentences, 43/43 is an in-sample regression verification rather than an out-of-sample adversarial estimate.

Table 7. Gauntlet suite (G1–G5 adversarial and G0 valid controls) across ablations and frozen v2.1 (V2) vs v3.1 (V3).

Stratum	Kind	Mechanism	Cases	V0	V1	V2	V2s	v2.1 full (V2)	v3.1 full (V3)
G1	Adversarial (rejected)	3-tuple cycles, 1-way/duplicate cross-clause thefts, comma-appositive swaps, although/though	8	0	0	8	8	8/8	8/8
G2	Adversarial (rejected)	Unicode Mn/Me combining & enclosing marks, non-ASCII Zs spaces, Cyrillic/Greek homoglyphs	8	0	5	8	8	6/8	8/8
G3	Adversarial (rejected)	Case-sensitive SI prefixes (uM vs um, mM vs mm) and designator collisions (Fig. 28 d)	8	0	2	8	8	4/8	8/8
G4	Adversarial (rejected)	Symbolic/never-negated inequalities (not < 5%, never > 28), bare/stacked negation	8	0	0	8	8	1/8	8/8
G5	Adversarial (rejected)	Symmetric Arm A and Arm B, parenthetical side-notes, and >= 3 arms in a pairwise clause	6	0	0	6	6	1/6	6/6
G0	Valid control (accepted)	Valid although contrasts, % before ./(, 28-d hyphens, colon-bearing PMID:/DOI: IDs	10	10	10	10	10	8/10	10/10

On the 48 cases of the gauntlet suite, frozen release v2.1 rejected 20/38 adversarial probes and accepted 8/10 valid controls: specifically, v2.1's _UNIT_CASEFOLD fallback silently folded lowercase length units (um, mm, nm, pm) into molarity (uM, mM, nM, pM), accepted Fig. 28 d as 28 days, failed to invert not < 5% and never > 28 days, permitted symmetric Arm A and Arm B without a contrast connector, and falsely rejected valid although multi-tuple sentences and colon-bearing PMID:/DOI: tuple IDs. Release v3.1 rejects 38/38 adversarial probes and accepts 10/10 valid controls.

6.7 Exhaustive finite-domain and execution-hygiene self-tests

Table 8. Exhaustive finite-domain, dual-mode, two-layer, and execution-hygiene self-tests.

Self-test	Result supported	Exhaustive domain	Outcome
Transfer classifier against a literal transcription of Definition 1	Theorem 1(1)	26,344 ordered context pairs (both vignette lattices)	0 disagreements
Search for a scalar ranking of the model_system axis	Theorem 1(2)	46,656 rank assignments (6 elements)	0 faithful; best assignment misclassifies 6 ordered pairs
Single-coordinate mutation of the true typed proposition	Theorem 3	19 atoms × 7 coordinates = 133 mutants	19/19 true propositions accepted; 133/133 mutants rejected
Equal-in-substance and loosely typed propositions	Theorem 3	2 reordered arm sets; 19 sample sizes given as floats	2/2 accepted; 19/19 rejected
Ledger tampering	Proposition 1(iii)	182 tampers of two ledgers (15 kinds)	0 accepted; 1/1 cross-study transposition accepted, as the proposition allows
Changes that replay does not bind	Proposition 1, remark	10 reworded rationales, renamed events and edit-and-revert pairs	10/10 accepted by replay; 10/10 change the ledger digest
Canonical rendering round trip	Corollary 2.1	242 atoms, 7 with direction words, digits or superscripts in names and units	242/242 renderings accepted by the strict lint
Mode A vs Mode B self-attesting adversary & non-prose surface locks	Invariant I6 & Theorem 2	1,161 record-level adversarial controls; 310 valid attested records; 60 non-prose metadata tampers	1,161/1,161 Mode A & 1,161/1,161 Mode B semantic rejections (0 escapes); 310/310 valid accepted; 60/60 metadata tampers rejected
Layer-1 structure-only closure vs Layer-2 digest locks & colon IDs	Proposition 1 & Invariants I1–I6	26 structural tampers (check_top_digests=False); 10 digest-only metadata tampers; 4 colon-ID, 5 arm-binding & 8 negated-bound vectors	26/26 structural tampers rejected without digests; 10/10 accepted by Layer 1 and 10/10 rejected by Layer 2; 4/4, 5/5 & 8/8 vectors passed
Source-pinned module loader & single-read JSON/package hygiene	Execution integrity	Poisoned PYTHONPYCACHEPREFIX .pyc (matching mtime + size); 2 loader tampers; 3 NaN/Infinity payloads; 13 manifest tampers	Poisoned .pyc bypassed (__cached__ is None); 2/2 loader tampers, 3/3 non-finite JSON & 13/13 hygiene tampers rejected

Across the 1,161 record-level adversarial cases outside the residual suite, Mode B self-attesting tampering achieves 1,161/1,161 semantic rejections (0 escapes), so in this benchmark an attestation hash never masks

a semantic weakness, while verify_non_prose_surface_lock rejects 60/60 metadata injections and edits outside prose_sentences. Two execution-hygiene points matter for anyone building a synthesis pipeline on this code. First, Python can load cached bytecode (a .pyc file) in place of the source file, so a tampered cache could silently replace the verifier. load_source_pinned_module avoids this by checking the SHA-256 of the .py source bytes and compiling those bytes directly with __cached__ = None, which defeats even a poisoned PYTHONPYCACHEPREFIX bytecode file crafted to match the source file's exact modification timestamp and byte length when -B is omitted. Second, zero assert statements appear anywhere in the live codebase, so python3 -O, which strips assertions, executes 100% of checks identically.

6.8 Primary-source real-text evaluation on peer-reviewed PubMed studies

To measure how CEG verification behaves beyond author-constructed synthetic templates, ceg_real_text_evaluation.py evaluates 12 peer-reviewed PubMed studies—four in idiopathic pulmonary fibrosis (INPULSIS-1/2 [33], ASCEND [34], BI 1015550 Phase 2 [35], and human IPF fibroblast MRTF inhibition [36]), four in heart failure (PARADIGM-HF [37], PARAGON-HF [38], VICTORIA [39], and post-MI rat LCZ696 remodeling [40]), and four in thoracic oncology (KEYNOTE-024 [41], AURA3 [42], ADAURA [43], and preclinical EGFR exon 20 insertion models [44]). Together these papers yield 31 EvidenceAtoms spanning point_level, relative_change, interval_bound, and censored_observation quantities over a five-axis biomedical context lattice (evidence_tier, indication, population_stratum, followup_horizon, readout_level) built by build_clinical_translational_lattice. The curation replay uses constructed defects, as in §5.2. Settled corpus D_4 (SHA-256 prefix 23586ef51fcd) is generated from the curated atoms, and initial corpus D_0 (SHA-256 prefix b762d8d68986) is that corpus with four defects written into it, of kinds that occur in extraction. The ledger's 4 reversible JSONPatch curation events repair them—one an OCR um → uM SI case fold on PMID:36585256, two an arm-swapped PARADIGM-HF primary event rate (the focal and the comparator value), and one an ITT-versus-CNS subgroup sample size (419 → 144) on AURA3—passing bidirectional I_1 binding and I_2 replay with zero errors. The replay is a conformance check and not a record of errors observed during extraction.

The evaluation has two tracks. Track A tests raw sentences copied verbatim from published PubMed abstracts. Track B tests structured multi-study syntheses and realistic clinical translation mistakes built over the same studies.

Track A: Verbatim PubMed abstract sentences ($n = 36$, 3 per study). Track A tests verbatim sentences extracted directly from the published PubMed XML abstracts both unanchored (raw journal sentences with no CEG tuple tag) and source-anchored (appending, in square brackets, the tag(s) of the study's curated outcome(s), such as R101:O1, with verbatim body text untouched).

Table 9a. Real-text Track A: verbatim PubMed abstract sentences ($n = 36$) across IPF, Heart Failure, and Thoracic Oncology.

Regime	Unanchored verbatim accepted	Source-anchored accepted	Source-anchored refused	IPF (n = 12)	HF (n = 12)	Thoracic Oncology (n = 12)
V0 citation presence	0/36 (0.0%)	36/36 (100.0%)	0/36 (0.0%)	12/12	12/12	12/12
V1 study-wide bag of numbers	0/36 (0.0%)	8/36 (22.2%)	28/36 (77.8%)	4/12	1/12	3/12
V2 prose lint, pooled numerals permitted	3/36 (8.3%)	7/36 (19.4%)	29/36 (80.6%)	4/12	1/12	2/12
V2S prose lint, pooled numerals rejected	3/36 (8.3%)	6/36 (16.7%)	30/36 (83.3%)	3/12	1/12	2/12
V3 full CEG	3/36 (8.3%)	6/36 (16.7%)	30/36 (83.3%)	3/12	1/12	2/12

For unanchored sentences the V3 row repeats the V2S result: a sentence without a tuple tag has no record for the full verifier to check. Source-anchored V3 accepts 6/36 (16.7%) verbatim abstract sentences and refuses 30/36 (83.3%). By the first violation raised, 21 sentences fail on numerals that no cited atom licenses (I3_TOK_UNGROUNDED_VALUE): inline 95% confidence/credible intervals, hazard/rate/odds ratios, ± standard errors, derived between-arm differences, patient event numerators/denominators (897 of 2526 patients) alongside the primary arm percentages, or trial-wide randomization and design numerals; 4 sentences, one in *Eur Respir J* and three in *N Engl J Med*, contain publisher XML non-ASCII spaces (a U+2005 four-per-em space in 50-150 µM, and U+2009 thin spaces around the equals sign of reported P values such as P = 0.06), caught as I3_TOK_FORBIDDEN_UNICODE; 3 sentences pool multiple outcomes in one clause (I3_CLAUSE_AMBIGUOUS_POOLED_CITATION); and 2 contain glued mutation tokens (A763insFQEA, H773insH), caught as I3_TOK_NONCANONICAL_NUMERAL. The five trial-

design sentences also name two trial groups without an explicit contrast connector (I3_CLAUSE_NON_CONTRAST_MULTI_ARM), as a further violation. The refusal codes do not separate the causes of refusal. Of the 30 refused sentences, 16 report findings that the study's curated atoms represent in full and are refused for the form of journal prose alone; 4 report a finding for which no atom was curated (TA-R102-3, TA-R201-3, TA-R202-1, TA-R304-1), so the appended tag names a different outcome of the same study; 5 combine a curated finding with one or more uncurated ones (TA-R104-1, TA-R202-2, TA-R203-3, TA-R204-2, TA-R301-3); and 5 are trial-design or randomization sentences (TA-R101-1, TA-R102-1, TA-R103-3, TA-R204-1, TA-R303-1). The 30/36 figure is therefore the refusal rate of verbatim abstract sentences against this 31-atom extraction, and an upper bound on what rewriting alone would cost. Track A demonstrates empirically that CEG prose checking is **not** a zero-shot verifier over raw published abstracts, meaning that it cannot be pointed at unedited journal prose and be expected to accept the true sentences: using CEGs requires expressing quantitative claims in tuple-scoped synthesis syntax or extending EvidenceAtom slots to carry confidence intervals and event counts.

Track B: Grounded clinical and translational synthesis records ($n = 84$). Track B evaluates 24 valid synthesis records (B-P1–B-P4) and 60 realistic clinical synthesis defects (D1–D6, 10 per family) built over the same 12 PubMed studies.

Table 9b. Real-text Track B: grounded synthesis records ($n = 84$: 24 valid, 60 adversarial) across 12 PubMed studies.

Stratum	Kind	Mechanism	Cases	V0	V1	V2	V2s	V3
B-P1	Valid (accepted)	Matched-context clinical RCT promotion with typed proposition	9	9	9	9	9	9
B-P2	Valid (accepted)	Preclinical or Phase 2 bridge behind an explicit falsifier gate	6	6	6	6	6	6
B-P3	Valid (accepted)	Adverse-event or neutral counter-evidence scoping record	5	5	5	5	5	5
B-P4	Valid (accepted)	Comparative, censored-bound, and although multi-trial syntax	4	4	4	4	4	4
D1	Adversarial (rejected)	Cross-trial pooling across distinct endpoints, regimens, or EF strata	10	0	1	1	6	10
D2	Adversarial (rejected)	Active-vs-comparator arm swap, appositive comma swap, non-contrast multi-arm	10	0	0	9	9	10
D3	Adversarial (rejected)	Biomarker/stage/CNS subgroup outcome promoted as overall ITT population	10	0	0	3	3	10
D4	Adversarial (rejected)	Rounded numeral, SI case fold (um vs uM), parenthesized unit or sample-size role swap	10	0	3	10	10	10
D5	Adversarial (rejected)	Fibroblast, rat, or PDX mouse outcome promoted directly to clinical context	10	0	0	0	0	10
D6	Adversarial (rejected)	3-trial cycle, 1-way theft, negated inequality (not <), Unicode Mn/homoglyph split	10	0	3	10	10	10

On Track B, V3 accepts 24/24 (100.0%) valid clinical synthesis records and rejects 60/60 (100.0%) adversarial synthesis defects—including 60/60 semantic rejections under the Mode B self-attesting adversary. By contrast, running frozen release v2.1 head-to-head on Track B accepts only 20/24 valid records (falsely rejecting two B-P2 translational bridges over the five-axis clinical lattice, one B-P3 safety-scoping record, and one B-P4 although contrast between two HFrEF trials) and rejects 50/60 adversarial records (letting 10 clinical synthesis defects escape: one in D1, four in D2, two in D4, and three in D6).

6.9 Production-scale case study: a type 1 diabetes corpus of 1,151 studies

In plain terms: the earlier sections test the checks on invented cases and on 12 curated studies. This section looks at what happens when the same representation is applied to a real corpus with roughly a hundred times as many studies, built for an engineering project on type 1 diabetes. It reports the corpus's size and structure, replays its correction log through the reference verifier, and says plainly what the corpus, as it stood in a dated snapshot, does not show. It is a case study and not a benchmark: no hypothesis is scored.

The corpus. The corpus comes from the OpenIslet project, which compiles studies on islet replacement and on immune-evasive stem-cell-derived islets (OpenIslet, openislet.org, 12 November 2026 snapshot). Every figure and status statement in this section describes that snapshot, and later states of the corpus may differ. The snapshot is not bundled inside this package and is identified by SHA-256 digests (Code and Data Availability). Of 3,922 screened records (3,921 distinct identifier pairs), 3,246 advanced at title-and-abstract screening, and 1,151 studies were extracted into structured records: 747 of the candidate records and all 404 advanced records of the project's earlier core list. The other 2,095 advanced records, a lower-priority

supporting catalog, were not extracted, so the corpus is a prioritized subset of the advanced literature and not a census of it. Screening, extraction, and verification were carried out by language models working as extractor, verifier, and adjudicator, with deterministic validation of each extraction round, as that project's runbook records; the models are named in the AI Contribution Statement. OpenIslet is an open-source project led by the author of this paper (Funding and Competing Interests). Ten findings from eight of its studies are the worked example of §5, where each is classified against a clinical target context and submitted to the gate. Table 10 summarizes the result.

Table 10. Production type 1 diabetes corpus (OpenIslet, 12 November 2026 snapshot): size, structure, and verification status.

Quantity	Value
Screening	3,922 screened records (3,921 distinct identifier pairs); 3,246 advanced; 523 parked as context; 153 excluded as off topic
Extraction	1,151 studies extracted (747 candidate records, 404 core-list records); 2,095 advanced records not extracted
Records	7,659 study arms; 14,105 outcomes
Links	2,447 study-to-design links, each tying one study to a component of the project's design: 1,029 supports; 784 scoping qualifications; 613 novel upgrade hypotheses; 19 noninformative; 2 contradicts
Studies by evidence tier	158 human clinical; 59 non-human primate; 50 humanized mouse; 296 human or stem-cell-derived islet in vitro; 562 rodent or cell line; 26 biophysical model or regulatory
Outcomes by direction	5,311 favorable; 3,262 context-dependent or mixed; 3,166 unfavorable or toxic; 1,827 null; 539 unresolvable
Outcome source	every outcome carries a verbatim quote, pinned to results text for 9,919 outcomes, to an abstract for 3,014 outcomes (21.4%), to a figure caption for 721, to a table for 363, and to methods text for 88
Reach	9,119 of 14,105 outcomes are cited by a study-to-design link or a dossier; 4,986 outcomes (35.3%) appear as study-arm background only
Primary-source status	0 of 14,105 outcomes certified against the primary source
Data-quality flags	4,949 outcomes (35.1%) need figure digitization; 8,580 outcomes (60.8%) are typed as qualitative; 726 of 1,151 studies have a PubMed Central identifier
Dossiers	407 indexed dossier records (378 distinct); 1,565 cited outcome tuples, of which 139 carry a point-level quantity and 1,426 are typed as missing numbers
Correction log	106 reversible events across 29 studies
Size	settled corpus 20,412,758 bytes in canonical JSON (17.7 kB per study; 1.4 kB per outcome) and 25,520,705 bytes as stored; 41,746,322 bytes of raw extraction packets; three single-window Markdown views of 9,644,648, 3,677,507, and 3,239,524 bytes (16,561,679 in total); full indexed graph 70,625,398 bytes

Size. The figures in the last row of Table 10 are byte counts and not token counts. In the canonical JSON form used in §7.3, the settled corpus occupies about 6.0 times as much per study as the 12-study corpus of §6.8. The difference lies in how many outcomes were extracted from each study and not in the size of a record: about 12.3 outcomes per study against about 2.6 outcomes per study, at 1.4 kB per outcome (arm and link records included) against 1.1 kB per outcome. The project also renders three Markdown views of the corpus, each intended to be read in a single context window, of 3.2 to 9.6 million bytes. At a rule of thumb of 3 to 4 bytes per token, which was not measured here, the smallest view is of the order of 0.8 to 1.1 million tokens, the largest of 2.4 to 3.2 million tokens, and the canonical settled corpus of 5.1 to 6.8 million tokens. Whether a view fits therefore depends on the context window of the model, and the complete corpus is several times larger than its smallest view. No model was run on any of these views in this paper.

Correction log at scale. The corpus's curation ledger holds 106 reversible events. Each is one edit to one study's record, with the old value, the new value, a reason, and digests of that study's record before and after: 25 events from the first to the second corpus version, 16 from the second to the third, and 65 from the third to the fourth. They edit outcome records (89 events), study-to-design links (9), arm lists (5), and study-design fields (3). Replaying the ledger forward from the first pinned corpus version and backward from the fourth reproduces the pinned payloads exactly, with no old-value guard or study-digest failure in either direction under the project's own digest convention. The reference implementation's replay verifier (invariant I_2, Proposition 1) then accepts the same ledger over the 1,151-study corpus in Layer 1 and in Layer 2, once the ledger is adapted to the corpus shape that the verifier expects and its study digests are recomputed in the canonical form of RFC 8785 (106 events, 29 patched studies). Because those digests are recomputed, Layer 2 acceptance of the unaltered ledger is expected by construction; the informative results are the exact replay in both directions, the check of the ledger's own stored digests, and the tamper controls. Eight kinds of tamper, run as ten controls, behave at this scale as Proposition 1 predicts. Layer 1 rejects 5 of 5 structural tampers

without consulting any digest: deleting an event, duplicating one, altering one string in either corpus, and redirecting a pointer. Transposing two adjacent events of one study that edit different fields, and overwriting one per-step digest, are accepted by Layer 1 and rejected by Layer 2. Rewording one rationale is accepted by replay but changes the ledger digest.

Dossiers and the project's own validator. The project compiled the corpus into bulk dossiers of cross-study failure modes, scoping qualifications, and upgrade hypotheses, written through rounds of language-model review. Their active records cite 1,565 outcome tuples across 320 studies, and 407 indexed dossier records (378 distinct identifiers) were checked by the project's own validator, which according to the snapshot's own record visited every active record (11,116 numeric-token occurrences checked, none unvisited). That validator is a separate implementation of the same design and is not part of this package. These counts are therefore reported for scale and are not results of the reference implementation. Of the 1,565 cited outcome tuples, 139 carry a point-level quantity and 1,426 are typed as missing numbers, so in this snapshot the typed layer (Theorem 3) can check a numeric magnitude for only a small fraction of the cited tuples; for the others it checks the remaining coordinates and the typed statement that no number is available. The dossiers' own verdicts authorize no numeric promotion: every upgrade candidate is held pending prospective falsification. That is the behavior the commit gate is designed to produce when no outcome has been certified against its primary source (invariant I_5).

What the case study shows and does not show. It shows that the representation and the verification of a correction log operate on a corpus of 1,151 studies and 14,105 outcomes, and it makes the limits of that corpus visible. It does not show that the extraction is accurate, because in this snapshot none of the outcomes has been certified against its primary source and 21.4% are pinned to an abstract rather than to full-text results. It does not show coverage, because 2,095 advanced records were not extracted. It does not show that a model can use such a corpus in context, because no model was run on it and the complete corpus is several times larger than its smallest single-window view. It does not show the quality of the hypotheses in the dossiers, because none is scored. Section 7.3 states these limits again among the limits of the whole evaluation.

7. Discussion

7.1 Exact guarantees versus heuristic lint boundaries

In plain terms: four parts of the design are exact (the structured claims, the edit log, the rule for carrying results between settings, and the metadata locks), and six things lie outside what the checks can guarantee. Both lists matter to anyone relying on a CEG.

Four properties are exact relative to the declared inputs: (1) a `TypedProposition` passing I_3^{typed} equals its target `EvidenceAtom` on all seven coordinates, and every promoted claim covers all supporting tuples (Theorem 3); (2) a corpus pair and ledger passing I_2 determine one another bijectively, and digest-guarded replay rejects all single-event deletions, same-study transpositions, pointer aliases, and cross-study pointer escapes, with replay-neutral metadata edits pinned by $H(\mathcal{L})$ in Layer 2 (Proposition 1); (3) direct quantitative promotion under I_4 licenses only monotone transfers $c_t \preccurlyeq_{\mathcal{C}} c_s$ on the declared product poset Λ, avoiding the inevitable false transfers or false exclusions of scalar rankings on incomparable models (Theorem 1); and (4) non-prose metadata surface locks (I_6) prevent unverified strings from bypassing verification outside `prose_sentences`, verified under a self-attesting Mode B adversary.

Six boundaries remain outside formal guarantees and must be stated plainly:

1. **Upstream extraction fidelity.** No invariant checks D_0 against the source PDF. If an extractor misreads a table cell into D_0 and no curation patch corrects it, I_1–I_6 certify coherence with the erroneous atom. The production corpus snapshot of §6.9 makes this limit concrete: its outcomes were extracted by language models, 21.4% of them are pinned to an abstract, and at the time of the snapshot none had been certified against its primary source.
2. **Specification of the context poset Λ.** Theorem 1 guarantees soundness *relative to* Λ. Omitting a relevant validity axis (such as dose level in the synthetic bundle of §5.4) collapses distinct clinical cohorts onto one context element; Λ is a scientific modeling commitment that must be peer-reviewed alongside the graph.

3. **Open-class free prose.** As Theorem 2 and R1–R4 (220/240 undetected) demonstrate, surface prose linting cannot resolve arbitrary endpoint noun phrases, unitless role swaps, or open-class rhetorical negations. Decision-making consumers must read $\mathcal{P}_{\text{typed}}$, treating Σ_{prose} as human-facing commentary constrained by a conservative lint.
4. **Derived cross-atom quantities and rich clinical syntax.** Track A (30/36 refusals on verbatim PubMed sentences) and P7 (0/50 accepted) indicate the rewrite cost: inline confidence intervals, hazard ratios paired with event rates, numerator/denominator counts, and cross-arm differences must currently either be decomposed into canonical tuple-scoped clauses or await typed derived-arithmetic extensions.
5. **Prospective falsifier gates.** Requiring a ProspectiveFalsifierGate on a BOUNDED_BRIDGE records the exact unresolved axes and a stated pass threshold, which the verifier checks for presence and does not parse; it does not empirically validate the bridge prior to running the experiment [31, 32].
6. **Screening coverage of structured strings.** Forbidden-Unicode screening covers prose sentences; atom identifiers, endpoint names, units, arm identifiers, interval strings, and provenance fields; falsifier-gate fields; record and proposition identifiers; and ledger event identifiers, study identifiers, pointers, and rationales. Homoglyph screening covers prose sentences, endpoint names, units, and arm identifiers. Strings outside these lists are not inspected. Auxiliary envelope metadata is bound by exact matching and by the Layer-2 digests. Declared study contexts, study-arm catalogues, arm aliases, source-arm mention lists, and the context partial order are graph configuration (ContextEvidenceGraph fields outside atoms): neither the five section digests nor a claim attestation covers them, so a package's digests do not authenticate them, and they must be reviewed and versioned together with the graph. Single-outcome canonical rendering (render_atom_sentence) likewise depends on the prose-compatibility conditions documented in §4.3.

7.2 Practical deployment in LLM drafting loops: explore free in context, verify at commit

The primary use of a CEG is to give researchers and language models a way to load a large structured literature into context, explore it freely, and generate hypotheses, and the experimental bridges that would test them, that stay tied to the recorded findings. Deterministic verification serves that use: it keeps the shared evidence graph clean as it grows, and it does not restrict exploration. The intended deployment follows the two regimes of §1 and Figure 2. In Regime 1 the model works over the in-context graph of EvidenceAtoms and ContextSignature coordinates without constraint; nothing in the CEG design lints or restricts its exploratory reasoning, intermediate drafts, or narrative discussion. In Regime 2 only candidate commits reach the verifier. Full V3 verification is deterministic, uses only the Python standard library, and calls no model, so a drafting agent can submit a candidate SynthesisClaimRecord, receive structured violation codes (I3_TOK_UNGROUNDED_VALUE, I3_CLAUSE_AMBIGUOUS_POOLED_CITATION, I4_MISSING_PROSPECTIVE_FALSIFIER_GATE, I4_INCOMPLETE_PROSPECTIVE_FALSIFIER_AXES), and repair the draft iteratively. No latency measurement is reported for this loop.

The commit gate is strict about context transfer rather than advisory. A SUPPORTS or CANDIDATE_UPGRADE record is rejected (I4_INADMISSIBLE_CONTEXT_TRANSFER) when any of its supporting tuples is an INADMISSIBLE_TRANSFER; a CANDIDATE_UPGRADE over a BOUNDED_BRIDGE passes only with a ProspectiveFalsifierGate covering every unresolved axis and is then held (HOLD_FOR_PROSPECTIVE_FALSIFIER) rather than promoted; and direct promotion requires DIRECT_MATCH on every supporting tuple. A cross-context observation that does not meet these conditions can still be recorded as a SCOPING_QUALIFICATION, which is how moderator explanations enter the graph.

The refusal rate measured on Track A (30/36 verbatim abstract sentences refused, §6.8) applies only to the evidence-anchor sentences that enter the commit gate as prose_sentences. It is not a cost on the model's free-form exploration, reasoning, or narrative discussion, which the verifier never sees. These statements describe design intent: the benchmarks of §6 evaluate the commit gate only, and no experiment in this paper measures exploration quality or discovery rates.

Two deployment safeguards follow directly from these findings: (i) reward signals or acceptance gates must require TypedProposition coverage rather than prose-lint passage alone, preventing models from drifting into R1–R4 paraphrases; and (ii) publisher text ingested from XML/HTML sources should normalize typographic spaces (U+00A0, U+2005, U+2009) explicitly during pre-ingestion curation, with JSONPatch events recording any normalization applied to quoted provenance.

7.3 Limits of the evaluation and what remains to be measured

The evaluation in §6 supports claims about the commit gate. It does not yet support claims about discovery, and five limits should be kept in view.

1. **Exploration is not evaluated.** No experiment places a CEG in a model's context and scores the hypotheses that the model produces. The dossiers of the production corpus (§6.9) were written through rounds of language-model review, but this paper does not score them: no recall, precision, or discovery rate is measured. Whether in-context exploration over a CEG finds real moderating variables, and how often its hypotheses survive testing, is the central open question.
2. **Size is measured, but fit in a context window is not.** Two corpora have been measured. In canonical JSON, the 31 `EvidenceAtom`s of §6.8 occupy 35,212 bytes in total, with a median of 1,155 bytes per atom (range 912 to 1,232). That is about 2.9 kB per study, with one to four curated outcomes per study. That density is not representative of production records: the settled production corpus of §6.9 averages 17.7 kB per study in the same canonical form, because about 12.3 outcomes were extracted per study, and its three single-window views occupy 3.2 to 9.6 million bytes each. The canonical form writes out every empty field and is not optimized for size. These figures describe record size only: the number of tokens depends on the model's tokenizer, no model was run with a large graph in its context, and how reasoning quality changes as the graph grows is not measured. At a rule of thumb of 3 to 4 bytes per token, the smallest view is of the order of 0.8 to 1.1 million tokens and the largest of 2.4 to 3.2 million tokens.
3. **The synthetic suites are in-sample.** The reference implementation was revised with access to every synthetic case (§6.1), so the synthetic results are regression checks and are not estimates of performance on unseen errors.
4. **The real-text evaluation is small and curated.** It covers 12 studies, 31 findings, 36 verbatim sentences, and 84 synthesis records. The package contains no extraction pipeline: these records were curated for this evaluation, and the production corpus was extracted by a separate pipeline. Extraction at scale will introduce errors, and the checks cannot see an extraction error that is consistent with the rest of the extracted data (§7.1, item 1).
5. **The production corpus is a prioritized subset that had not been certified at the time of the snapshot.** In this snapshot it holds 1,151 of the 3,246 records that advanced at screening, and 2,095 lower-priority records were not extracted. Its records were extracted and verified by language models; 3,014 outcomes (21.4%) are pinned to an abstract rather than to full-text results, 4,949 outcomes (35.1%) need figure digitization, 8,580 outcomes (60.8%) are typed as qualitative, and none of the 14,105 outcomes is certified against its primary source. Its dossier records were checked by the project's own validator and not by the reference implementation, and only 139 of its 1,565 cited outcome tuples carry a point-level quantity. The same holds for the ten findings of §5: their quoted sentences were located in publisher-deposited text, but none is certified against a primary PDF, so no exact quantitative promotion is available in that bundle either.

8. Conclusion

Context–Evidence Graphs give researchers and language models a way to load a large structured literature into context, explore it freely, and generate hypotheses and experimental bridges that stay tied to the recorded findings. Verification serves that discovery work rather than restricting it: exploration is unconstrained, and a deterministic gate checks only what is written back, so the shared evidence graph stays clean as it grows. The gate is needed because common errors in automated quantitative synthesis recombine real citations and real numbers across incompatible outcomes, arms, units, bounds, and experimental contexts. CEGs separate what can be guaranteed deterministically (exact seven-coordinate typed propositions, invertible state-guarded curation ledgers, partial-order transfer gating across incomparable models, and non-prose metadata surface locks) from heuristic prose linting, whose impossibility bounds are proved and whose verbatim clinical refusal rates are measured explicitly. This paper formalizes the representation and benchmarks that integrity layer, and a production corpus of 1,151 type 1 diabetes studies shows the representation and its correction log operating at that scale. Measuring how much of such a corpus a model can use in its context window, and how well exploration over a CEG yields sound hypotheses, is the next step.

Code and Data Availability

The zero-dependency reference implementation (`ceg_reference.py v3.1`), domain vignettes (`domain_vignettes.py`), synthetic benchmark (`ceg_synthetic_benchmark.py` and `benchmark_results.json`), 12-study PubMed real-text evaluation (`ceg_real_text_evaluation.py` and `real_text_results.json`), and primary bibliography ledger (`audit/BIBLIOGRAPHY-PRIMARY-LEDGER.json`) are included in the standalone companion package (https://github.com/openislet/ceg; overview and guide at https://openislet.org/ceg) and require only the Python standard library. Deterministic byte-identical JSON regeneration has been verified across CPython `3.9`–`3.14` under both normal and `-O` execution. The type 1 diabetes worked example of §5 is built by `domain_vignettes.build_t1d_vignette`, which also holds the synthetic `NX-410` bundle used by the benchmark; `t1d/check_t1d_vignette.py` rebuilds it and compares every finding with its record in the OpenIslet corpus, while its `--quotes` option locates each quoted sentence in the article text held by the National Center for Biotechnology Information (stored outputs `t1d/t1d_vignette_verification.json` and `t1d/t1d_vignette_quote_check.json`). The production-corpus figures of §6.9 are produced from the OpenIslet snapshot files by `t1d/t1d_corpus_metrics.py` and `t1d/replay_production_ledger.py`, whose `--recompute` option recomputes them except for the counts reported by that project's own validator and its certification status, which are read from the snapshot's own records; their stored outputs (`t1d/t1d_corpus_metrics.json` and `t1d/production_ledger_replay.json`) are included. A public projection of the settled 1,151-study type 1 diabetes Context–Evidence Graph is available under CC BY 4.0 in the OpenIslet repository (https://github.com/openislet/openislet; https://openislet.org/blueprint). In it each verbatim quote is replaced by its SHA-256 digest, and text fields that overlap the source are replaced by structured descriptors, so the figures of §6.9 are computed from the staging files described next and cannot be recomputed from the public projection. The pre-export staging masters and curation ledger read by `--recompute` are not bundled in this package: the first and fourth master files are pinned by SHA-256 digests beginning 6f14f1ec627e and cb24687ac7a6, and the curation ledger by one beginning f28a2a22af7f (read-only staging archive available upon request from kev@openislet.org, with the stated SHA-256 digests verifying byte identity on retrieval). The frozen pre-hardening baselines behind the comparisons of §6.1, Tables 6 and 7, and §6.8 (the first release, `v2.0` and `v2.1`), and the scripts that compare releases, are held in the author's development archive and are available upon request from kev@openislet.org.

AI Contribution Statement

The following generative AI systems were used under the author's direction across this work and the November 2026 corpus snapshot of §6.9: ChatGPT-6 Astra, ChatGPT-6.1 Sol, Claude 5.5 Sonnet, Claude Fable 5.2, Claude Science, Gemini 4 Argon, Gemini 4 Carbon, and Muse Wonder Full (together with additional Claude and Gemini models). AI systems are not listed as authors; the author directed the work and takes full responsibility for all formal claims, code, and empirical results.

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Competing interests: The production corpus of §6.9, from which the studies of the worked example of §5 are drawn, was compiled by OpenIslet (openislet.org), an open-source research project led by the author. The author declares no financial or commercial competing interests.

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